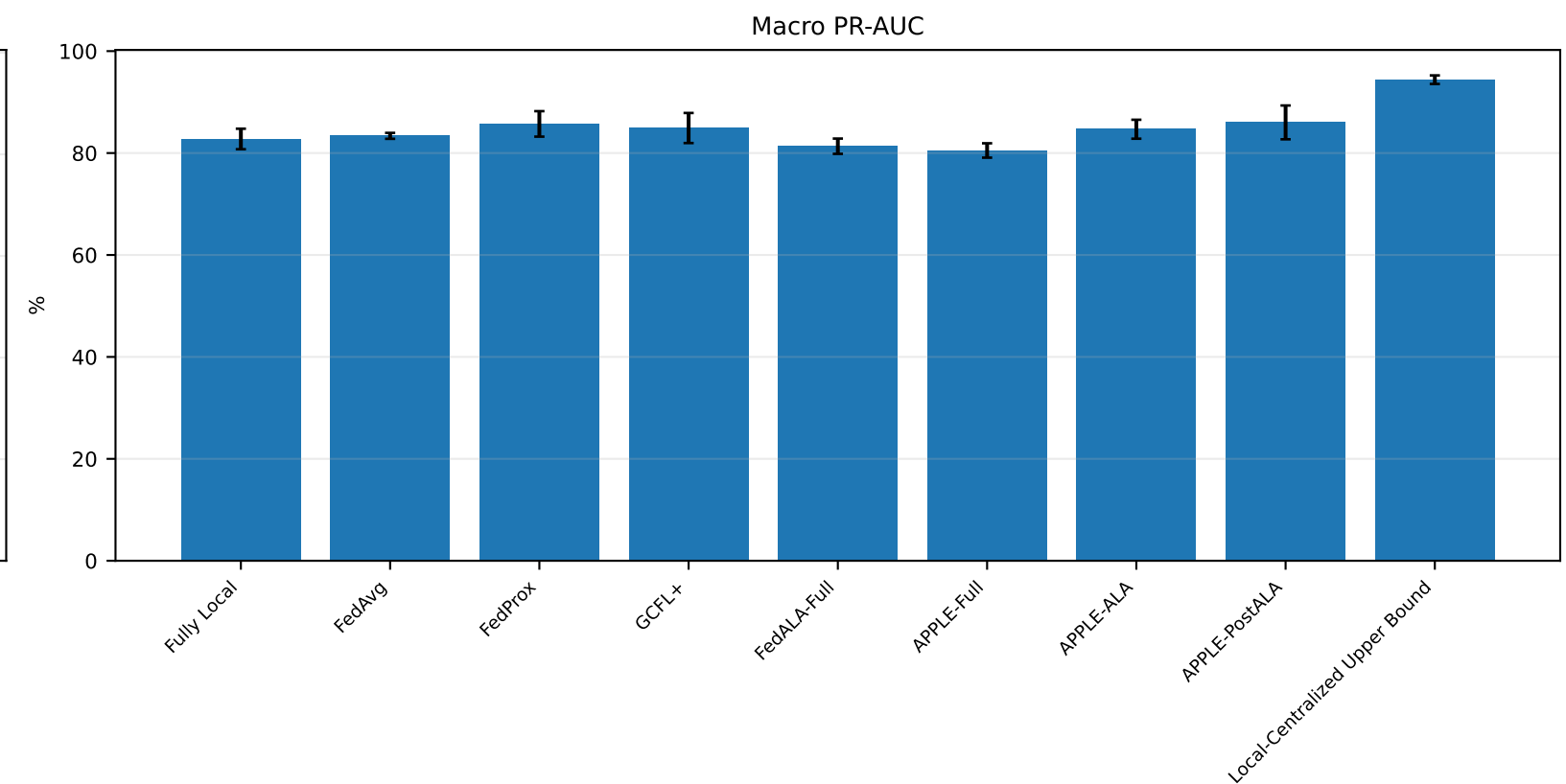
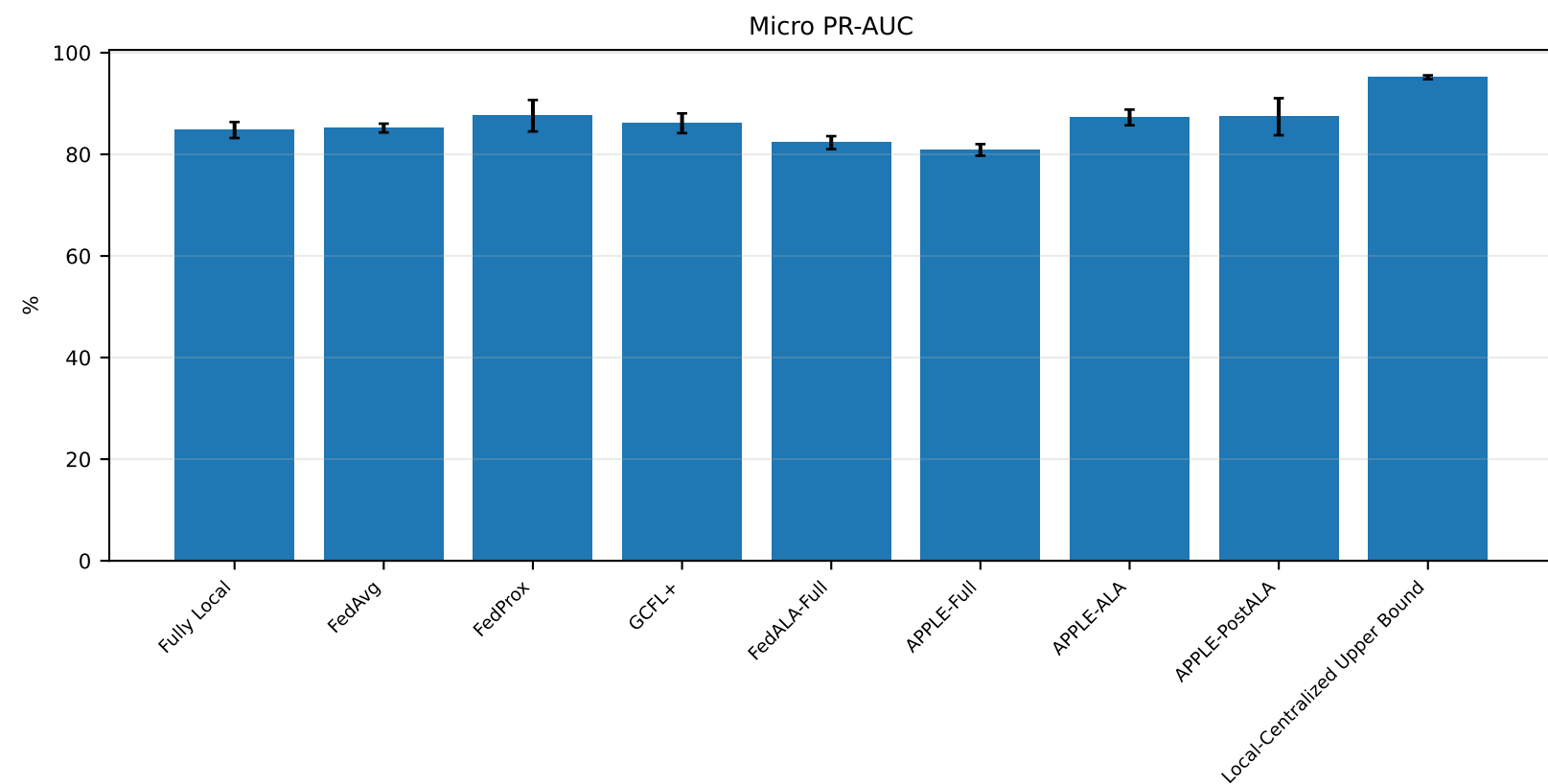
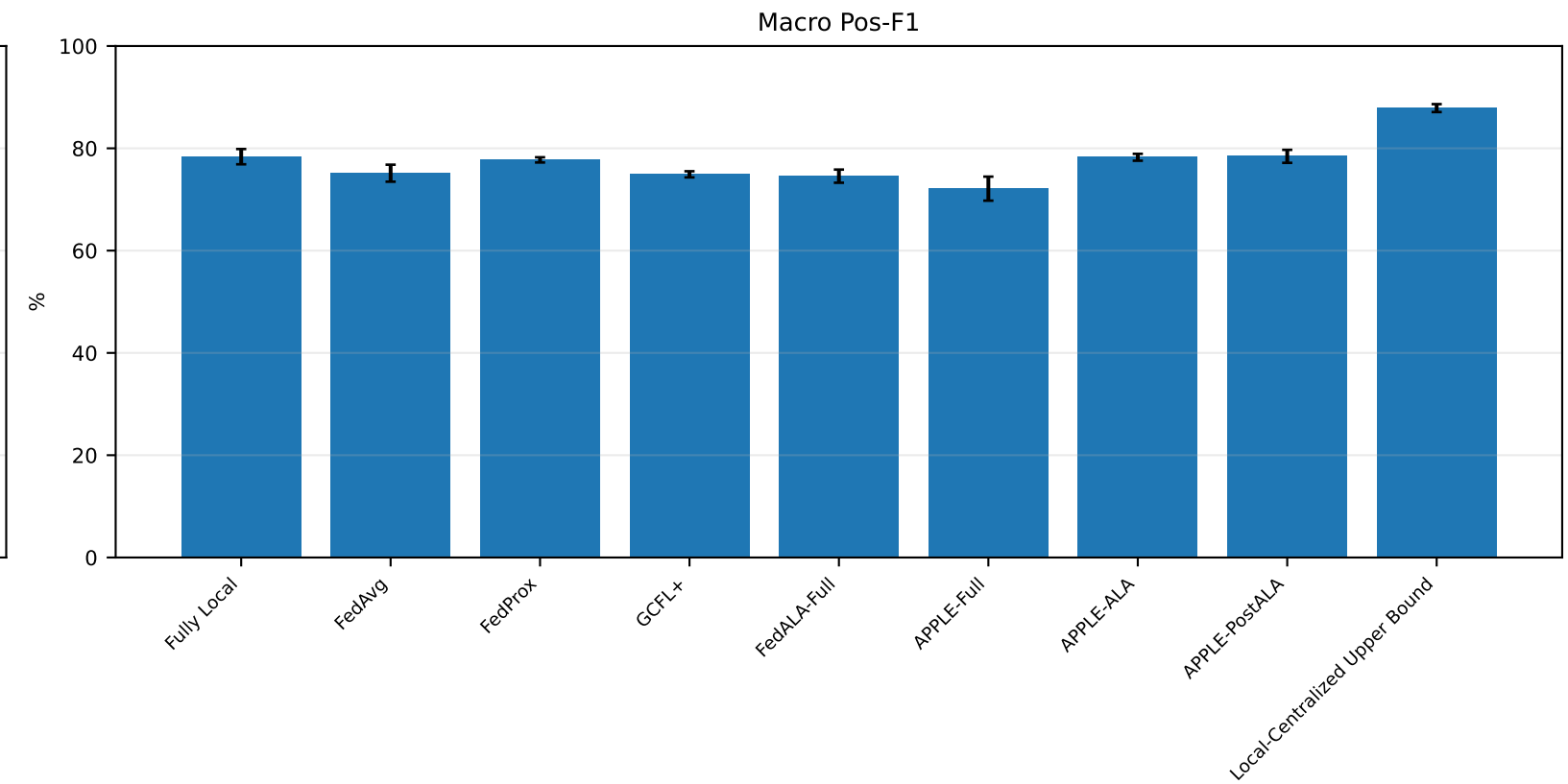
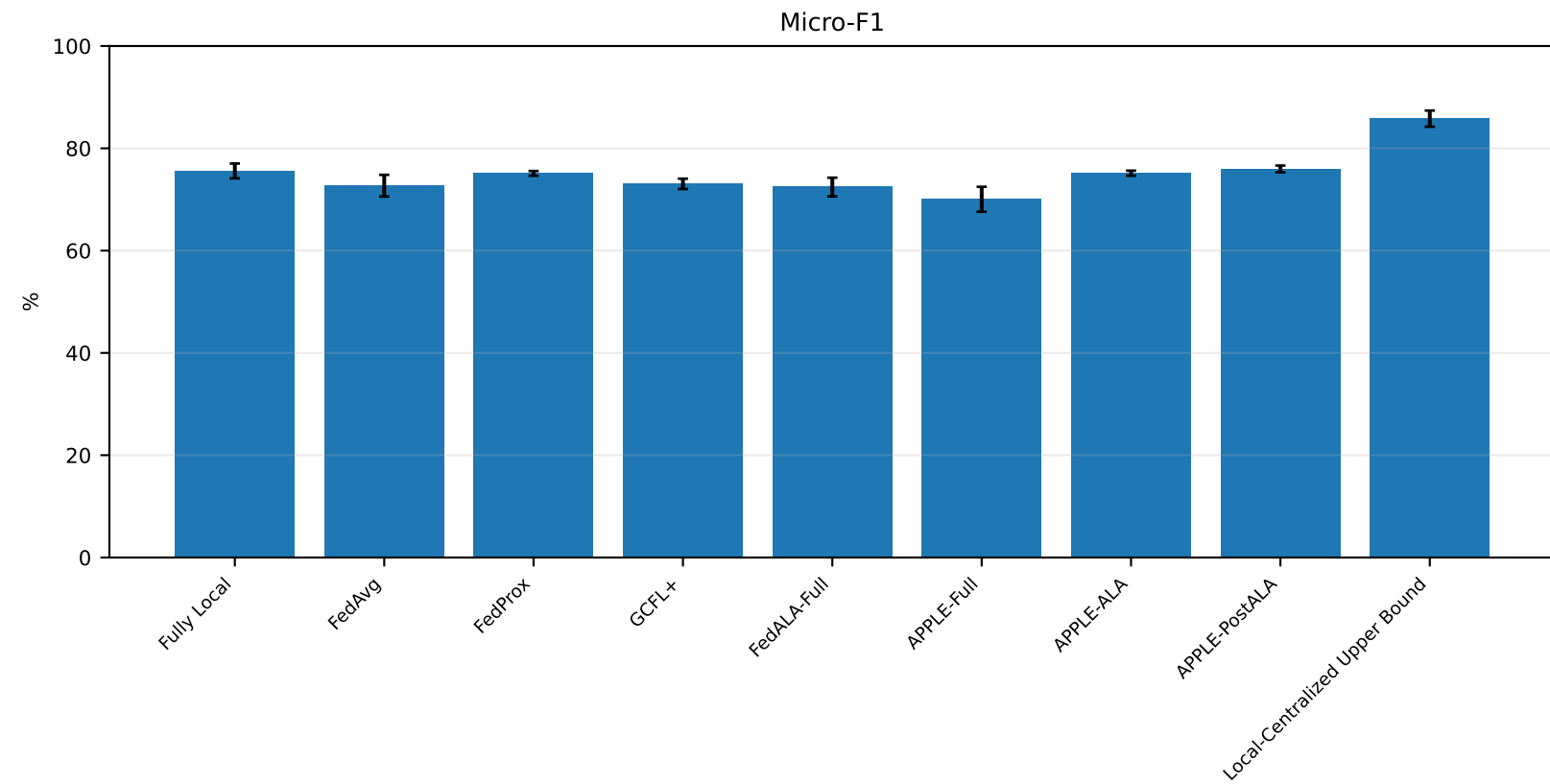


Main baselines + APPLE/FedALA/PostALA | IID / low heterogeneity (q=0.2) | selector=micro_f1 | protocol=OO
O+O: oracle/full validation selection → oracle/full test

Method	Micro-F1	Macro Pos-F1	Micro PR-AUC	Macro PR-AUC	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1	Sel. round	n
Fully Local	75.6% ± 1.5%	78.4% ± 1.5%	84.8% ± 1.6%	82.8% ± 2.0%	92.1% ± 1.8%	87.2% ± 1.9%	78.9% ± 2.6%	76.3% ± 1.3%	57.4% ± 1.5%	—	3
FedAvg	72.7% ± 2.1%	75.1% ± 1.7%	85.2% ± 0.9%	83.4% ± 0.6%	88.7% ± 4.0%	83.1% ± 3.5%	74.2% ± 1.2%	69.6% ± 2.8%	60.1% ± 3.1%	75.7 ± 1.5	3
FedProx	75.1% ± 0.5%	77.7% ± 0.5%	87.6% ± 3.1%	85.7% ± 2.5%	90.8% ± 2.9%	87.9% ± 1.8%	77.1% ± 1.8%	71.7% ± 0.9%	61.2% ± 2.8%	74.3 ± 4.0	3
GCFL+	73.1% ± 1.0%	74.9% ± 0.6%	86.1% ± 1.9%	84.9% ± 3.0%	81.7% ± 4.3%	86.0% ± 0.5%	74.1% ± 1.5%	71.5% ± 3.6%	61.2% ± 3.1%	73.7 ± 3.1	3
FedALA-Full	72.4% ± 1.8%	74.6% ± 1.3%	82.3% ± 1.3%	81.3% ± 1.5%	85.3% ± 3.2%	85.3% ± 3.4%	72.7% ± 1.0%	69.7% ± 2.2%	59.6% ± 3.6%	63.3 ± 6.4	3
APPLE-Full	70.0% ± 2.4%	72.1% ± 2.3%	80.9% ± 1.1%	80.5% ± 1.4%	87.8% ± 1.0%	74.7% ± 6.5%	70.5% ± 1.2%	68.1% ± 3.3%	59.4% ± 2.2%	70.3 ± 13.3	3
APPLE-ALA	75.1% ± 0.5%	78.2% ± 0.7%	87.3% ± 1.5%	84.7% ± 1.9%	92.5% ± 2.2%	89.4% ± 1.2%	79.6% ± 2.6%	71.4% ± 1.4%	58.3% ± 0.6%	71.7 ± 9.7	3
APPLE-PostALA	76.0% ± 0.7%	78.4% ± 1.3%	87.4% ± 3.6%	86.0% ± 3.3%	89.1% ± 1.3%	89.7% ± 3.4%	78.8% ± 3.5%	73.6% ± 1.4%	61.1% ± 2.5%	70.3 ± 9.3	3
Local-Centralized Upper Bound	85.8% ± 1.6%	87.9% ± 0.8%	95.2% ± 0.4%	94.4% ± 0.8%	96.7% ± 1.6%	95.7% ± 1.2%	90.7% ± 2.8%	85.2% ± 2.6%	71.0% ± 5.8%	—	3

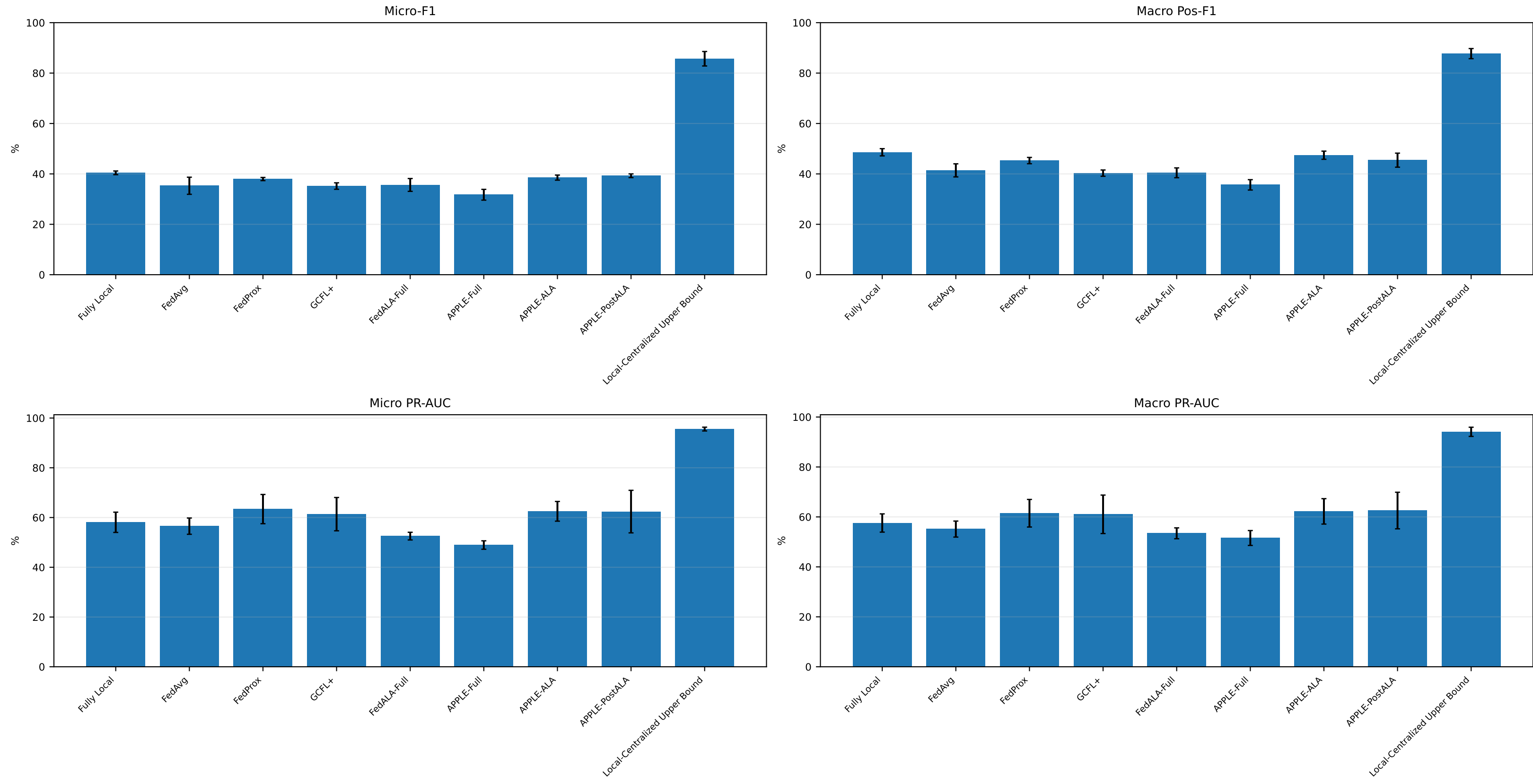
Main baselines + APPLE/FedALA/PostALA | IID / low heterogeneity ($q=0.2$) | selector=micro_f1 | protocol=OO | overview bars
O+O: oracle/full validation selection → oracle/full test



Main baselines + APPLE/FedALA/PostALA | IID / low heterogeneity (q=0.2) | selector=micro_f1 | protocol=VV
V+V: visible validation selection → visible test

Method	Micro-F1	Macro Pos-F1	Micro PR-AUC	Macro PR-AUC	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1	Sel. round	n
Fully Local	40.4% ± 0.7%	48.6% ± 1.4%	58.1% ± 4.1%	57.6% ± 3.7%	71.6% ± 2.5%	62.9% ± 1.8%	45.6% ± 2.4%	40.4% ± 0.9%	22.5% ± 0.1%	—	3
FedAvg	35.3% ± 3.4%	41.4% ± 2.6%	56.5% ± 3.3%	55.1% ± 3.2%	64.1% ± 5.4%	51.6% ± 9.2%	36.2% ± 2.1%	31.5% ± 3.2%	23.7% ± 3.1%	70.3 ± 8.1	3
FedProx	38.0% ± 0.6%	45.3% ± 1.2%	63.4% ± 5.9%	61.5% ± 5.5%	67.5% ± 6.9%	60.4% ± 2.3%	40.5% ± 2.2%	33.8% ± 1.3%	24.2% ± 2.0%	74.3 ± 4.0	3
GCFL+	35.2% ± 1.3%	40.3% ± 1.2%	61.4% ± 6.7%	61.0% ± 7.7%	52.4% ± 1.5%	56.7% ± 5.1%	35.0% ± 1.5%	34.1% ± 4.1%	23.3% ± 1.0%	73.3 ± 3.1	3
FedALA-Full	35.6% ± 2.5%	40.4% ± 1.9%	52.5% ± 1.5%	53.4% ± 2.2%	56.2% ± 5.6%	54.7% ± 7.3%	34.9% ± 0.6%	32.5% ± 2.2%	23.9% ± 3.2%	61.0 ± 10.4	3
APPLE-Full	31.7% ± 2.1%	35.7% ± 2.1%	48.9% ± 1.7%	51.6% ± 3.0%	55.8% ± 3.5%	36.9% ± 7.7%	33.2% ± 2.3%	29.7% ± 2.6%	22.7% ± 1.0%	65.0 ± 11.8	3
APPLE-ALA	38.5% ± 1.0%	47.4% ± 1.6%	62.5% ± 4.0%	62.2% ± 5.1%	72.9% ± 3.1%	63.4% ± 4.1%	43.9% ± 2.1%	34.6% ± 2.5%	22.3% ± 0.3%	68.7 ± 5.0	3
APPLE-PostALA	39.3% ± 0.7%	45.4% ± 2.8%	62.4% ± 8.6%	62.6% ± 7.3%	60.9% ± 5.5%	61.4% ± 6.9%	44.6% ± 3.4%	36.0% ± 1.1%	24.4% ± 3.0%	77.0 ± 3.6	3
Local-Centralized Upper Bound	85.7% ± 2.9%	87.7% ± 2.0%	95.6% ± 0.8%	94.1% ± 1.8%	96.1% ± 1.1%	95.4% ± 1.1%	91.1% ± 2.1%	84.7% ± 4.9%	71.3% ± 6.2%	—	3

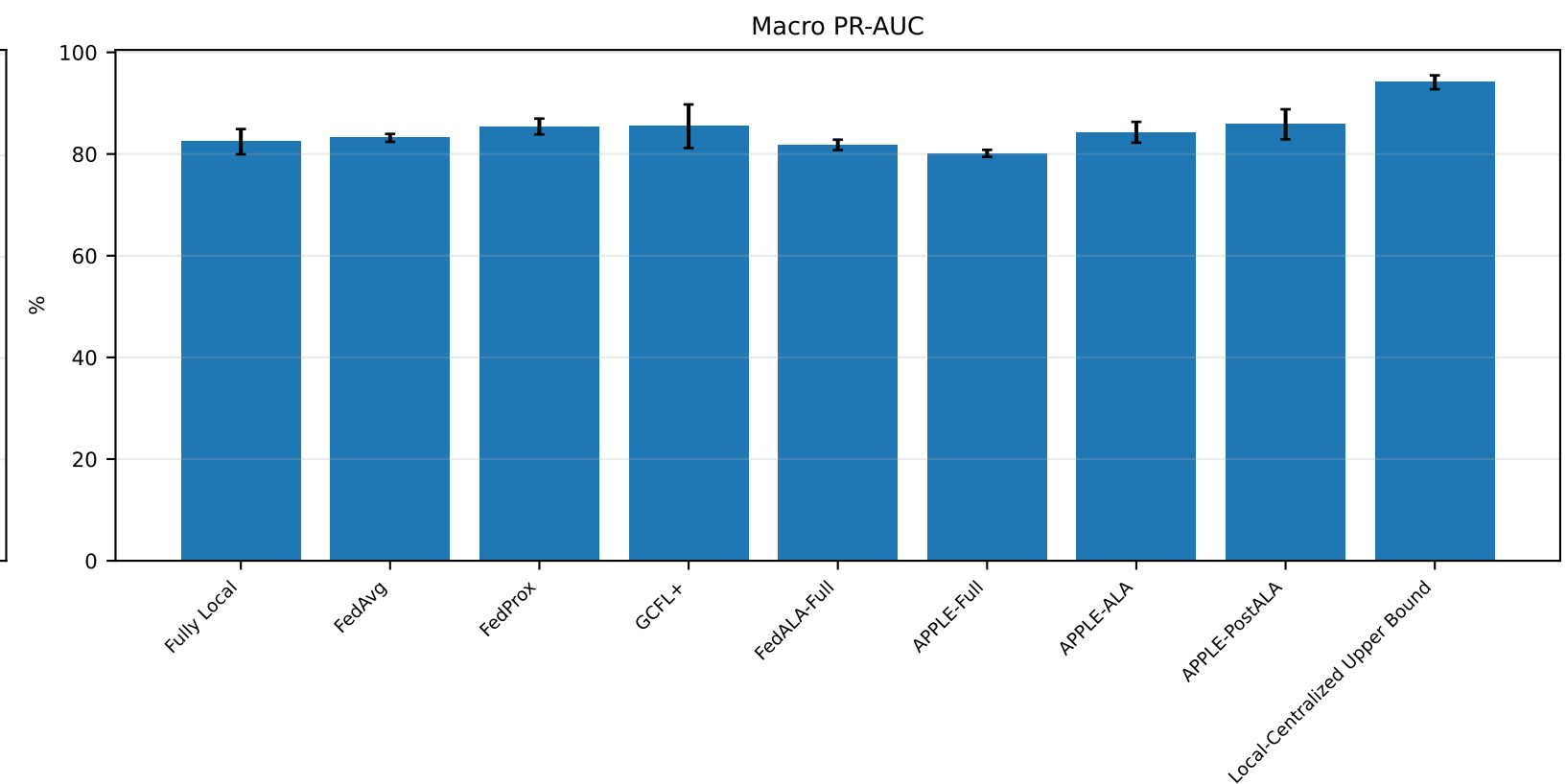
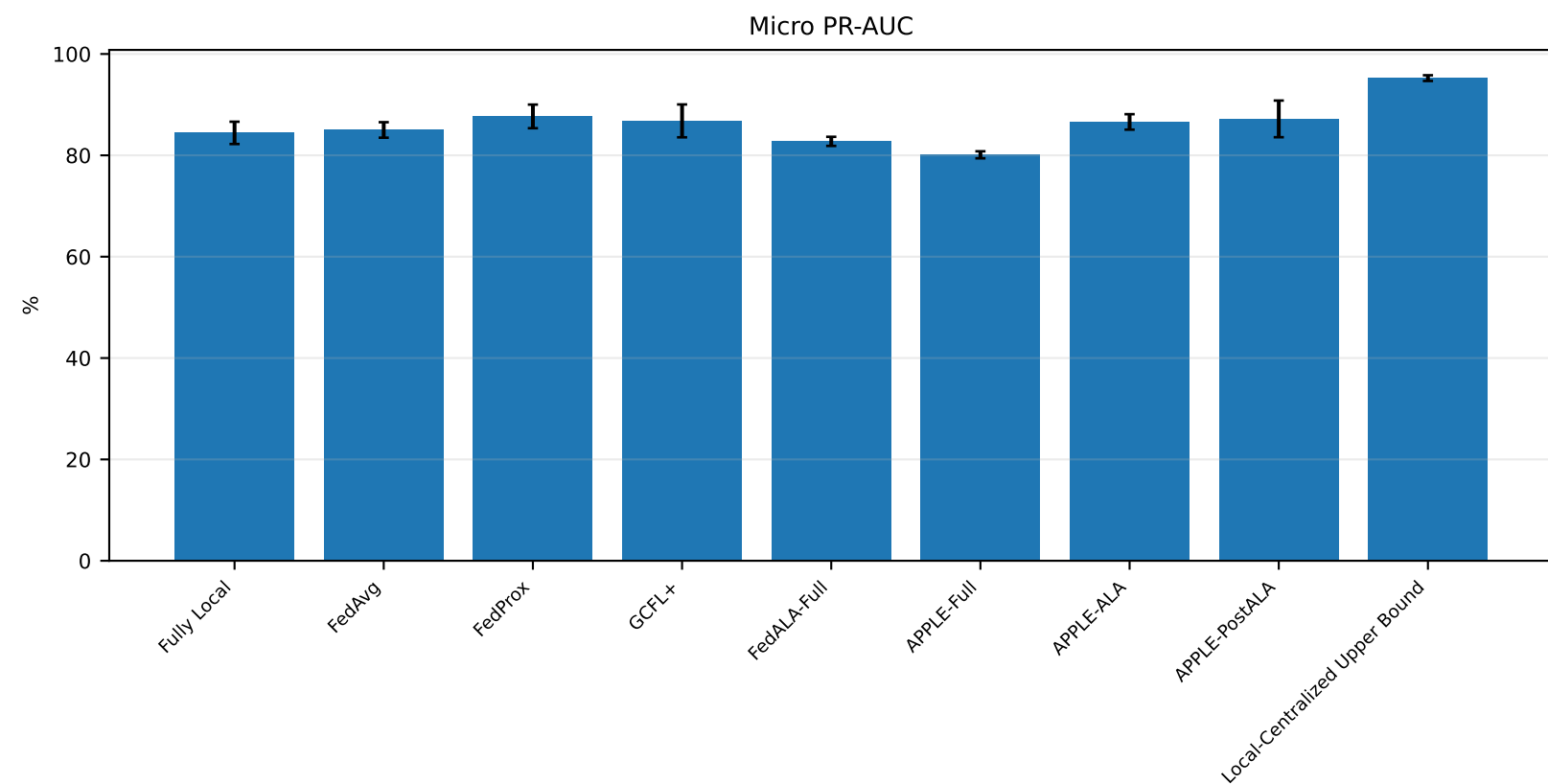
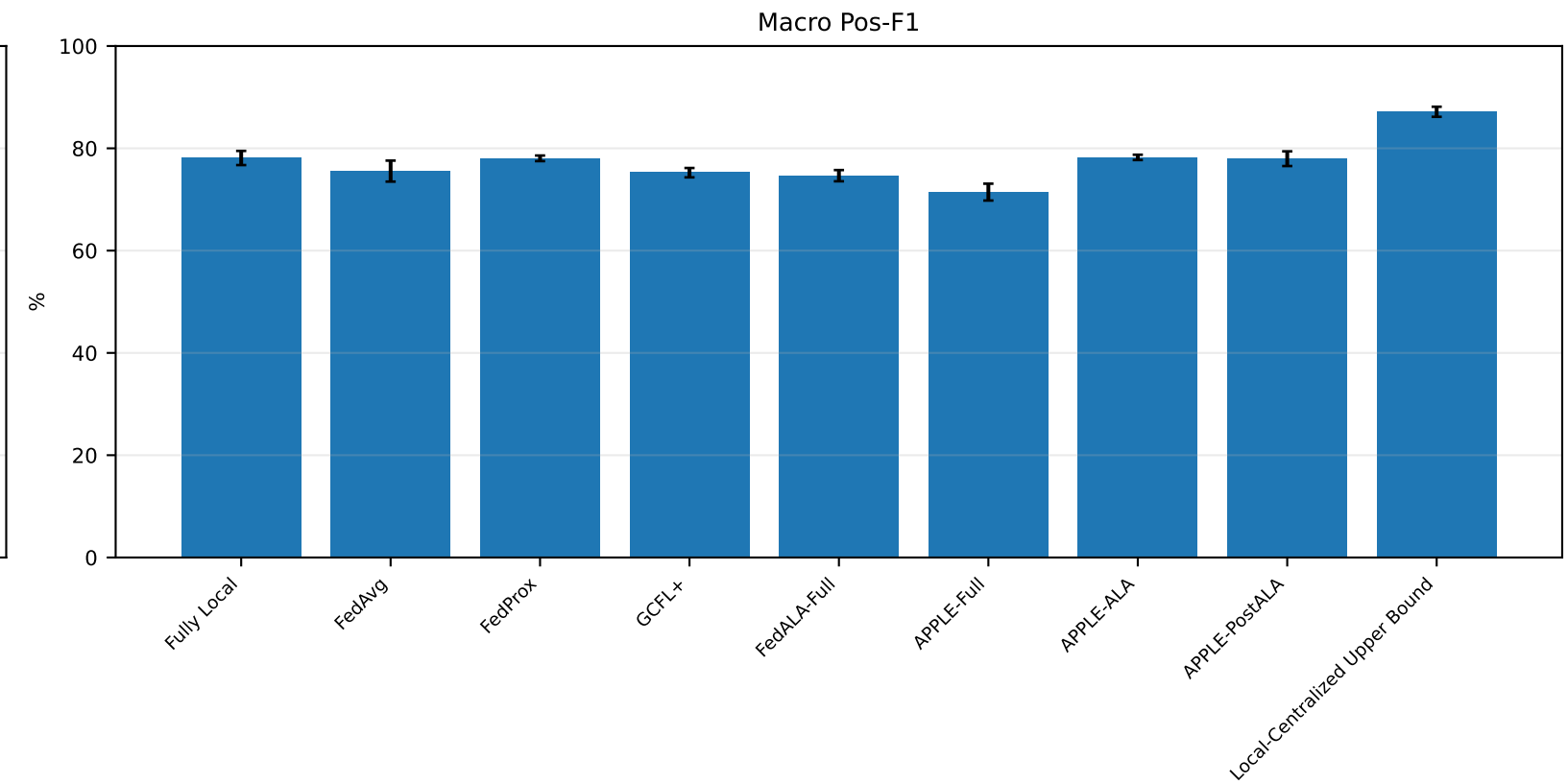
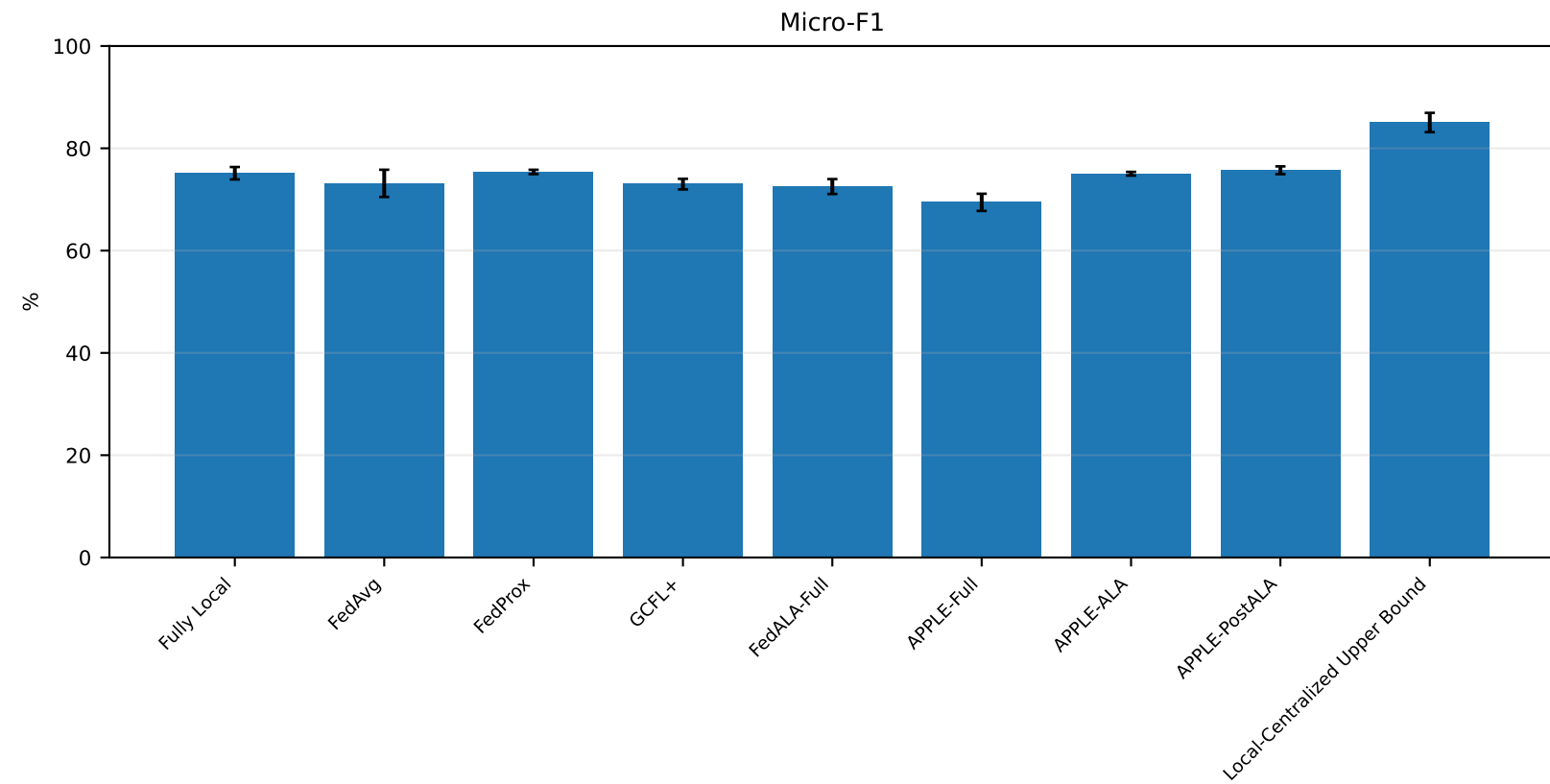
Main baselines + APPLE/FedALA/PostALA | IID / low heterogeneity ($q=0.2$) | selector=micro_f1 | protocol=VV | overview bars
V+V: visible validation selection → visible test



Main baselines + APPLE/FedALA/PostALA | IID / low heterogeneity (q=0.2) | selector=micro_f1 | protocol=VO
V+O: visible validation selection → oracle/full test

Method	Micro-F1	Macro Pos-F1	Micro PR-AUC	Macro PR-AUC	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1	Sel. round	n
Fully Local	75.1% ± 1.2%	78.1% ± 1.4%	84.4% ± 2.2%	82.4% ± 2.5%	91.6% ± 0.7%	88.3% ± 2.5%	78.5% ± 1.7%	76.1% ± 2.3%	56.0% ± 1.3%	—	3
FedAvg	73.2% ± 2.7%	75.5% ± 2.1%	85.0% ± 1.5%	83.2% ± 0.8%	89.1% ± 3.1%	83.8% ± 4.5%	73.7% ± 1.1%	70.0% ± 3.3%	61.0% ± 4.2%	70.3 ± 8.1	3
FedProx	75.4% ± 0.4%	78.0% ± 0.5%	87.7% ± 2.3%	85.4% ± 1.6%	90.7% ± 3.1%	88.8% ± 0.8%	77.0% ± 2.6%	72.4% ± 1.6%	61.3% ± 3.1%	74.3 ± 4.0	3
GCFL+	73.0% ± 1.0%	75.2% ± 0.9%	86.8% ± 3.2%	85.5% ± 4.3%	84.4% ± 0.8%	86.8% ± 1.4%	72.6% ± 1.6%	72.3% ± 3.8%	60.1% ± 1.5%	73.3 ± 3.1	3
FedALA-Full	72.5% ± 1.5%	74.7% ± 1.1%	82.8% ± 0.9%	81.8% ± 1.0%	86.0% ± 3.8%	85.0% ± 3.7%	71.9% ± 0.5%	70.3% ± 1.6%	60.0% ± 3.2%	61.0 ± 10.4	3
APPLE-Full	69.4% ± 1.7%	71.4% ± 1.6%	80.1% ± 0.7%	80.1% ± 0.7%	86.1% ± 2.4%	74.1% ± 5.5%	70.8% ± 2.1%	67.6% ± 2.5%	58.5% ± 0.7%	65.0 ± 11.8	3
APPLE-ALA	75.0% ± 0.4%	78.2% ± 0.5%	86.6% ± 1.5%	84.3% ± 2.0%	93.1% ± 0.6%	89.1% ± 1.8%	79.2% ± 1.2%	71.8% ± 1.9%	58.0% ± 0.7%	68.7 ± 5.0	3
APPLE-PostALA	75.7% ± 0.8%	78.0% ± 1.4%	87.2% ± 3.6%	85.8% ± 3.0%	87.7% ± 3.0%	88.6% ± 3.0%	78.8% ± 3.0%	73.5% ± 1.7%	61.2% ± 4.0%	77.0 ± 3.6	3
Local-Centralized Upper Bound	85.0% ± 1.9%	87.1% ± 1.0%	95.2% ± 0.6%	94.1% ± 1.4%	95.5% ± 1.3%	95.8% ± 2.3%	90.2% ± 1.7%	83.9% ± 4.3%	70.4% ± 5.7%	—	3

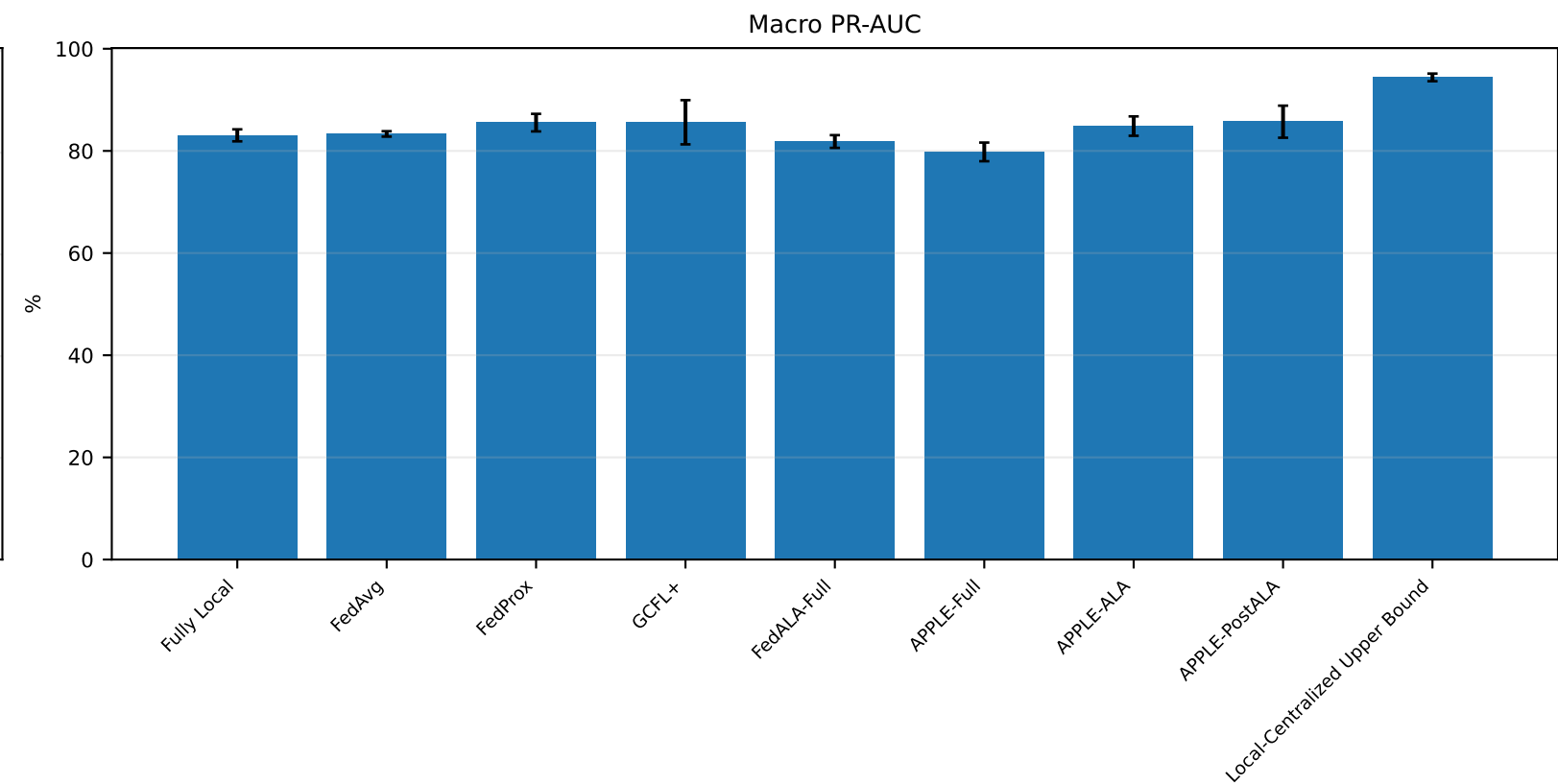
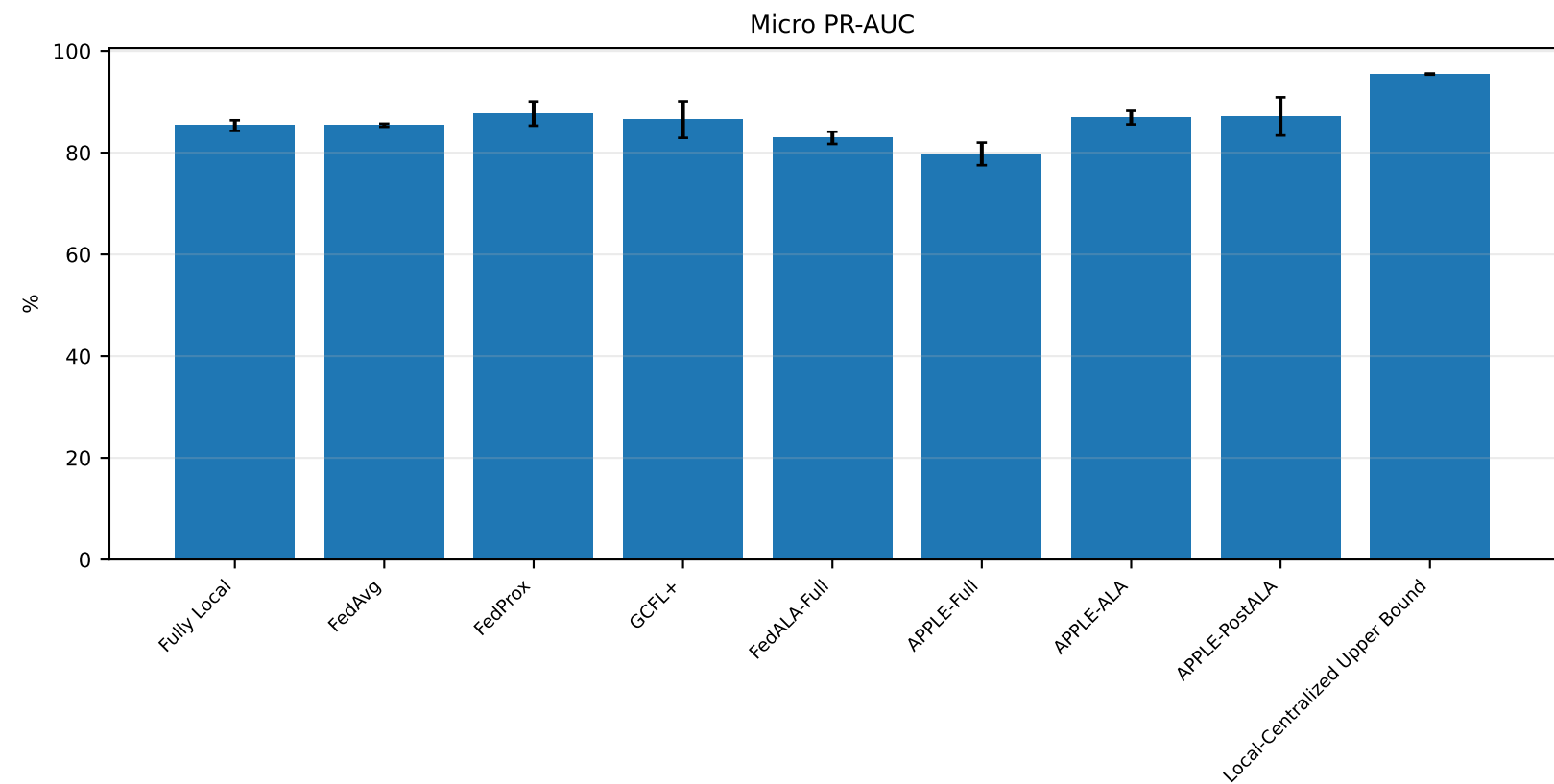
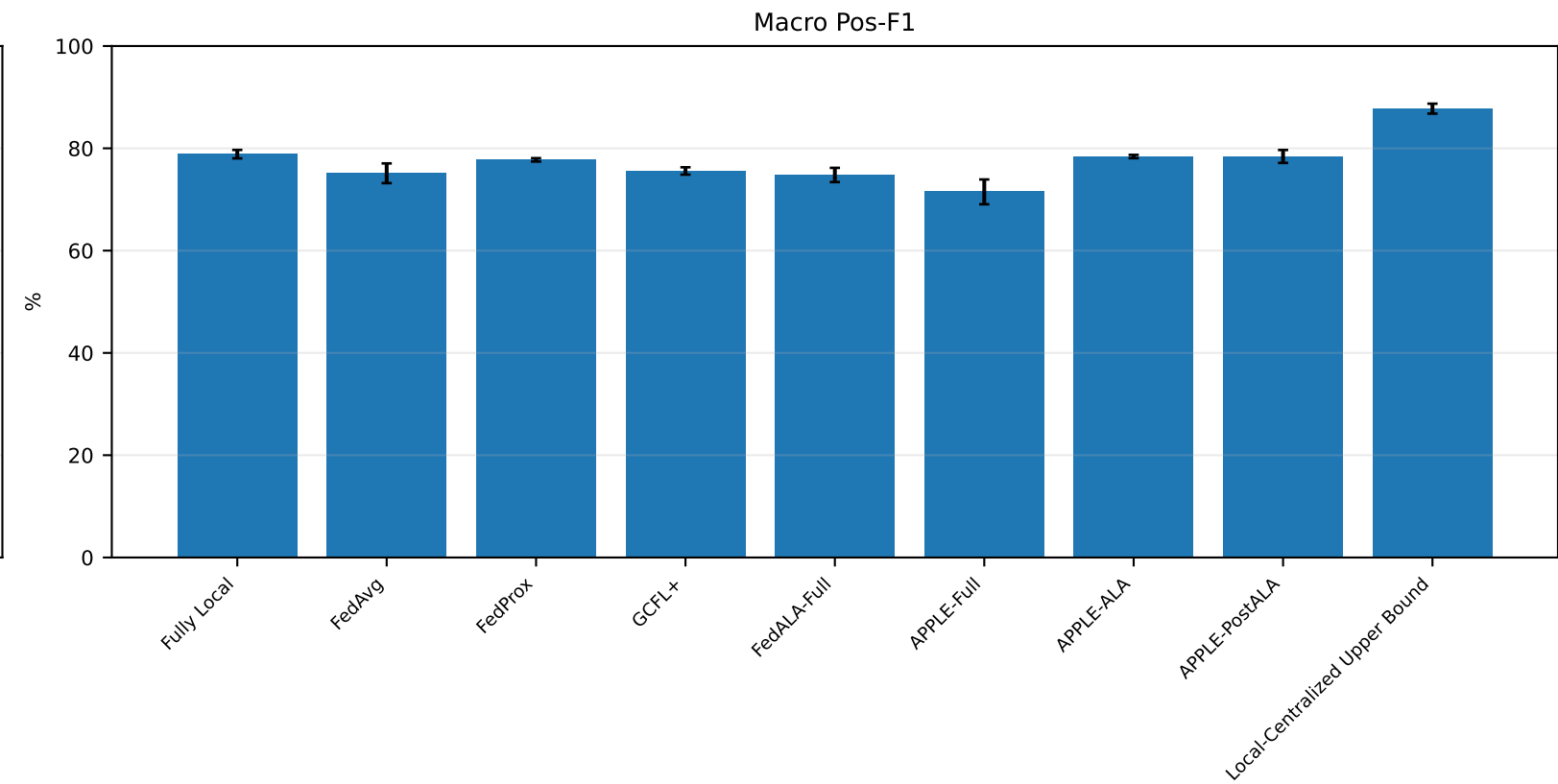
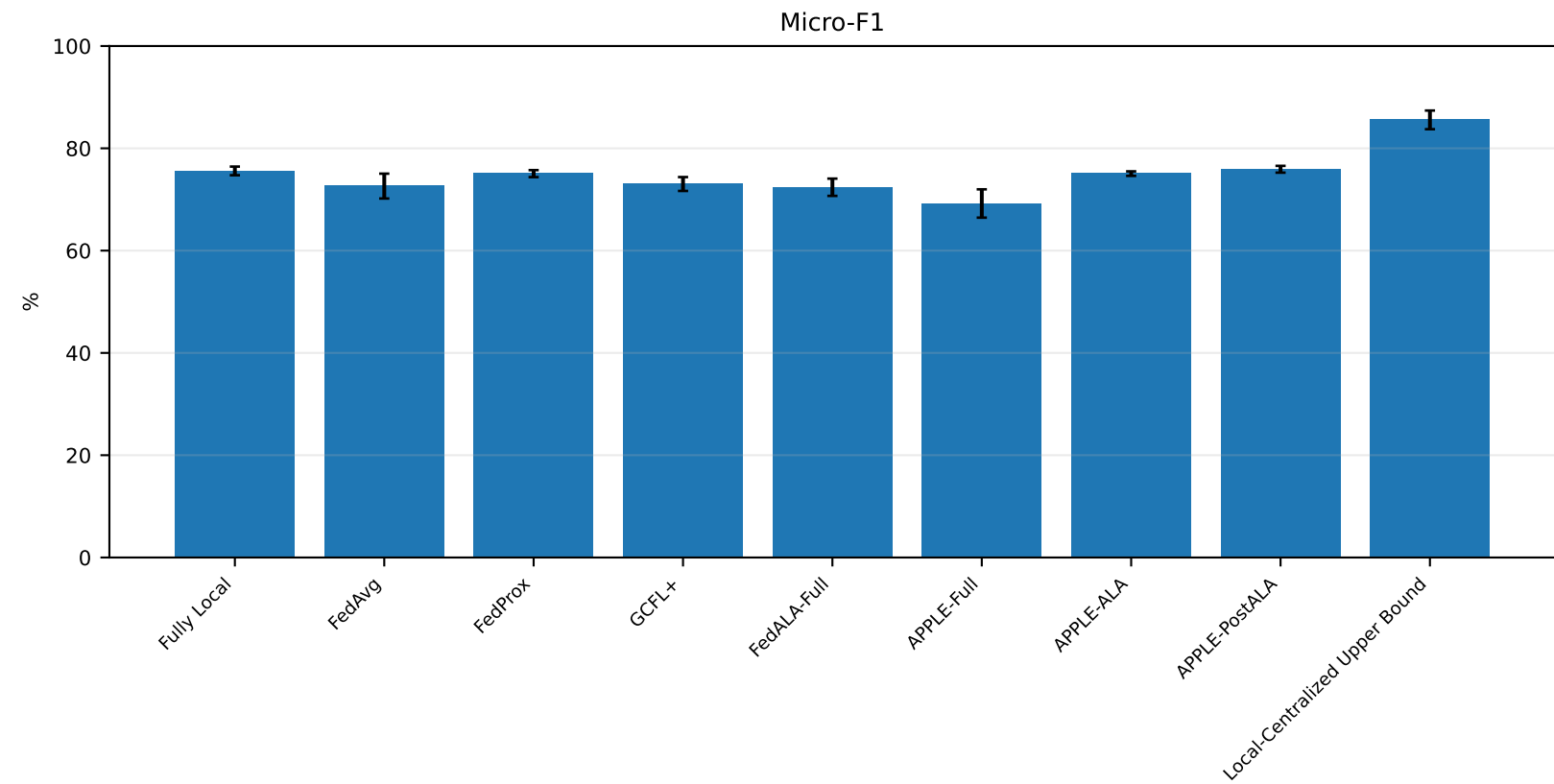
Main baselines + APPLE/FedALA/PostALA | IID / low heterogeneity ($q=0.2$) | selector=micro_f1 | protocol=VO | overview bars
V+O: visible validation selection → oracle/full test



Main baselines + APPLE/FedALA/PostALA | IID / low heterogeneity (q=0.2) | selector=macro_pos_f1 | protocol=OO
O+O: oracle/full validation selection → oracle/full test

Method	Micro-F1	Macro Pos-F1	Micro PR-AUC	Macro PR-AUC	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1	Sel. round	n
Fully Local	75.6% ± 0.8%	78.9% ± 0.8%	85.3% ± 1.0%	83.0% ± 1.2%	93.6% ± 0.9%	88.8% ± 1.1%	79.3% ± 2.4%	76.4% ± 0.4%	56.2% ± 0.6%	—	3
FedAvg	72.6% ± 2.4%	75.1% ± 1.9%	85.4% ± 0.3%	83.3% ± 0.5%	89.5% ± 3.5%	83.0% ± 3.9%	73.7% ± 1.6%	69.4% ± 3.2%	60.1% ± 3.3%	75.7 ± 1.5	3
FedProx	75.1% ± 0.7%	77.7% ± 0.3%	87.7% ± 2.4%	85.5% ± 1.7%	91.6% ± 3.0%	87.5% ± 1.7%	77.0% ± 1.4%	71.2% ± 1.8%	61.4% ± 2.4%	73.7 ± 4.7	3
GCFL+	73.0% ± 1.4%	75.6% ± 0.7%	86.5% ± 3.6%	85.6% ± 4.3%	86.3% ± 4.2%	87.5% ± 1.4%	72.6% ± 0.8%	71.9% ± 4.7%	59.6% ± 2.5%	74.7 ± 4.2	3
FedALA-Full	72.4% ± 1.7%	74.8% ± 1.4%	82.9% ± 1.2%	81.8% ± 1.2%	87.3% ± 1.7%	85.3% ± 3.7%	72.2% ± 0.7%	69.9% ± 2.0%	59.3% ± 3.0%	67.0 ± 0.0	3
APPLE-Full	69.2% ± 2.8%	71.5% ± 2.4%	79.8% ± 2.2%	79.8% ± 1.8%	87.3% ± 1.5%	77.4% ± 4.1%	67.8% ± 2.1%	66.5% ± 4.5%	58.4% ± 3.3%	76.7 ± 2.3	3
APPLE-ALA	75.1% ± 0.4%	78.4% ± 0.3%	86.9% ± 1.3%	84.9% ± 1.9%	92.3% ± 1.7%	89.6% ± 1.8%	81.0% ± 0.9%	71.5% ± 1.7%	57.6% ± 0.7%	75.0 ± 4.6	3
APPLE-PostALA	75.9% ± 0.7%	78.4% ± 1.3%	87.1% ± 3.8%	85.7% ± 3.1%	89.8% ± 1.4%	88.9% ± 2.5%	78.9% ± 3.8%	73.6% ± 1.4%	60.9% ± 2.9%	70.3 ± 9.3	3
Local-Centralized Upper Bound	85.6% ± 1.8%	87.8% ± 1.0%	95.5% ± 0.1%	94.4% ± 0.7%	97.0% ± 1.9%	96.0% ± 0.8%	90.2% ± 3.3%	85.1% ± 2.1%	70.4% ± 5.8%	—	3

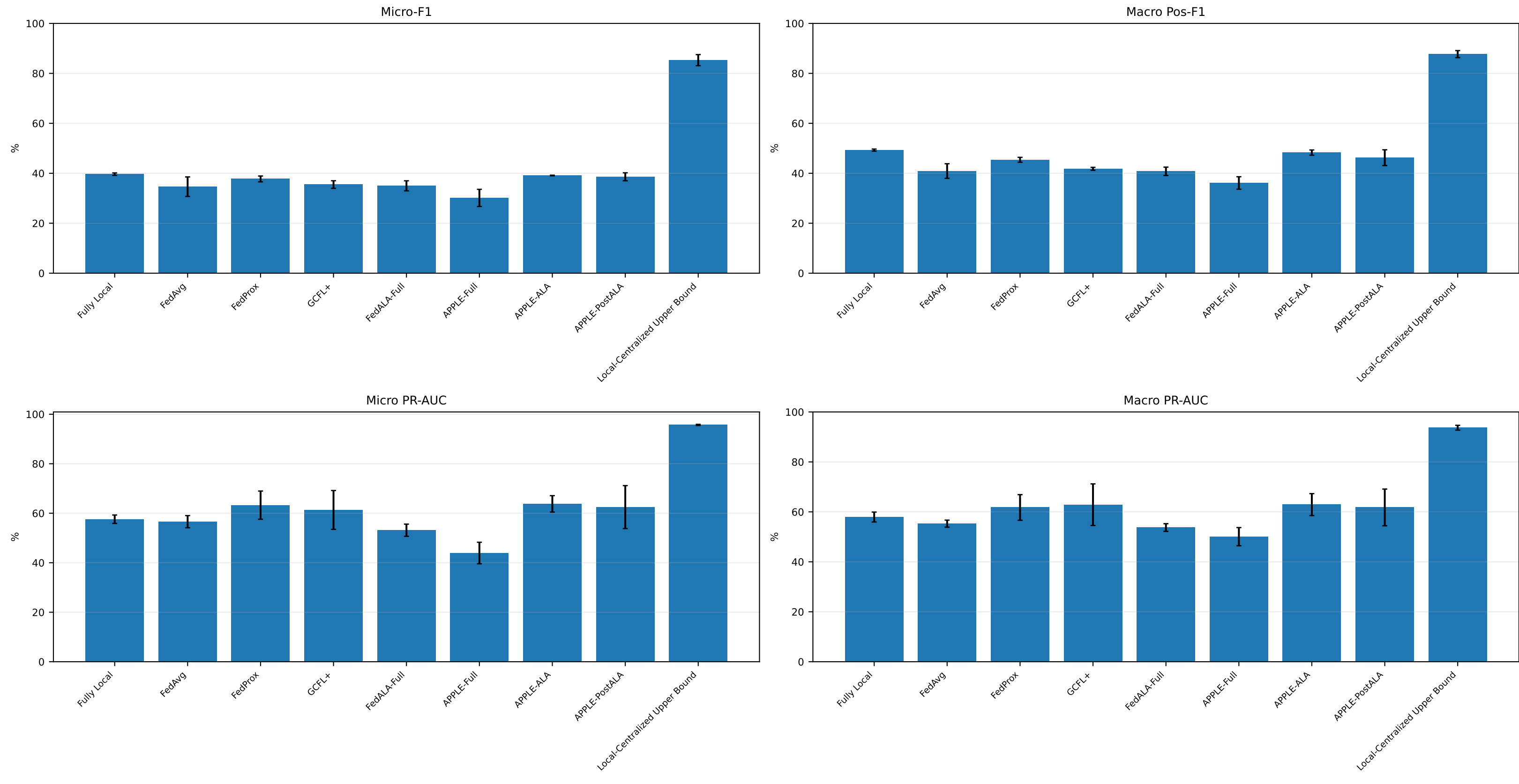
Main baselines + APPLE/FedALA/PostALA | IID / low heterogeneity ($q=0.2$) | selector=macro_pos_f1 | protocol=OO | overview bars
O+O: oracle/full validation selection → oracle/full test



Main baselines + APPLE/FedALA/PostALA | IID / low heterogeneity (q=0.2) | selector=macro_pos_f1 | protocol=VV
V+V: visible validation selection → visible test

Method	Micro-F1	Macro Pos-F1	Micro PR-AUC	Macro PR-AUC	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1	Sel. round	n
Fully Local	39.7% ± 0.5%	49.3% ± 0.4%	57.6% ± 1.7%	57.9% ± 2.0%	78.2% ± 2.5%	64.6% ± 2.1%	42.0% ± 1.8%	40.1% ± 0.7%	21.6% ± 0.0%	—	3
FedAvg	34.6% ± 3.9%	40.9% ± 2.9%	56.6% ± 2.5%	55.3% ± 1.4%	63.7% ± 8.7%	51.2% ± 9.0%	35.7% ± 1.7%	30.7% ± 4.7%	23.2% ± 3.7%	71.3 ± 9.0	3
FedProx	37.7% ± 1.2%	45.4% ± 1.0%	63.3% ± 5.7%	61.8% ± 5.1%	69.1% ± 8.0%	60.4% ± 2.2%	40.1% ± 2.7%	33.3% ± 2.8%	24.1% ± 2.2%	73.7 ± 4.7	3
GCFL+	35.5% ± 1.5%	41.8% ± 0.6%	61.3% ± 7.8%	62.9% ± 8.3%	56.8% ± 6.7%	60.1% ± 2.4%	34.5% ± 0.8%	34.5% ± 5.3%	23.1% ± 1.6%	74.7 ± 4.2	3
FedALA-Full	35.0% ± 2.0%	40.8% ± 1.7%	53.1% ± 2.5%	53.7% ± 1.5%	56.5% ± 4.1%	57.1% ± 5.3%	36.5% ± 2.4%	30.9% ± 1.7%	22.9% ± 2.4%	68.7 ± 7.6	3
APPLE-Full	30.1% ± 3.4%	36.1% ± 2.5%	43.9% ± 4.3%	50.1% ± 3.6%	62.6% ± 6.1%	39.9% ± 11.0%	29.0% ± 2.7%	28.2% ± 4.5%	21.0% ± 3.3%	76.7 ± 1.5	3
APPLE-ALA	39.1% ± 0.1%	48.3% ± 1.0%	63.8% ± 3.3%	62.9% ± 4.4%	74.6% ± 1.8%	63.5% ± 5.1%	45.7% ± 1.4%	35.2% ± 2.4%	22.4% ± 0.5%	72.7 ± 8.1	3
APPLE-PostALA	38.6% ± 1.6%	46.3% ± 3.2%	62.5% ± 8.7%	61.8% ± 7.4%	67.1% ± 11.5%	63.9% ± 7.8%	41.8% ± 5.4%	34.5% ± 2.1%	24.0% ± 3.6%	74.0 ± 8.7	3
Local-Centralized Upper Bound	85.3% ± 2.2%	87.7% ± 1.4%	95.7% ± 0.2%	93.7% ± 1.0%	97.5% ± 0.7%	97.7% ± 1.4%	90.6% ± 1.6%	83.0% ± 2.9%	69.7% ± 5.6%	—	3

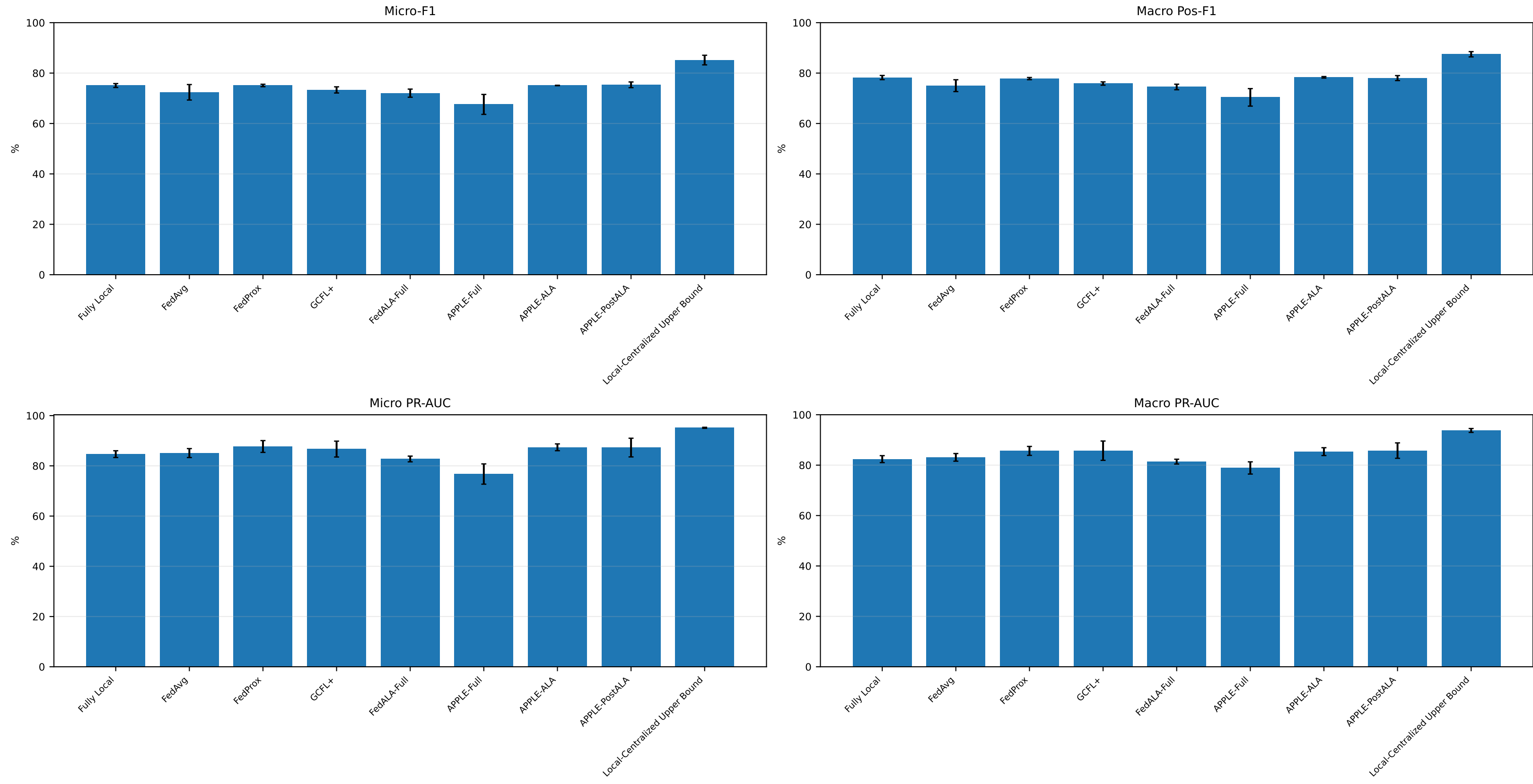
Main baselines + APPLE/FedALA/PostALA | IID / low heterogeneity ($q=0.2$) | selector=macro_pos_f1 | protocol=VV | overview bars
V+V: visible validation selection → visible test



Main baselines + APPLE/FedALA/PostALA | IID / low heterogeneity (q=0.2) | selector=macro_pos_f1 | protocol=VO
V+O: visible validation selection → oracle/full test

Method	Micro-F1	Macro Pos-F1	Micro PR-AUC	Macro PR-AUC	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1	Sel. round	n
Fully Local	75.1% ± 0.8%	78.2% ± 0.8%	84.7% ± 1.3%	82.4% ± 1.4%	93.7% ± 1.0%	88.3% ± 2.3%	77.3% ± 0.5%	76.2% ± 1.2%	55.7% ± 1.2%	—	3
FedAvg	72.4% ± 3.1%	75.0% ± 2.3%	85.1% ± 1.8%	83.1% ± 1.5%	89.9% ± 3.4%	83.2% ± 4.0%	73.2% ± 1.3%	68.6% ± 4.2%	60.1% ± 4.7%	71.3 ± 9.0	3
FedProx	75.1% ± 0.5%	77.8% ± 0.4%	87.7% ± 2.4%	85.7% ± 1.8%	91.7% ± 2.7%	87.9% ± 0.7%	76.8% ± 2.3%	71.2% ± 2.0%	61.5% ± 2.3%	73.7 ± 4.7	3
GCFL+	73.3% ± 1.2%	75.9% ± 0.7%	86.7% ± 3.2%	85.7% ± 3.8%	86.5% ± 3.3%	87.8% ± 0.9%	72.6% ± 1.1%	72.4% ± 4.8%	59.9% ± 2.1%	74.7 ± 4.2	3
FedALA-Full	72.0% ± 1.6%	74.5% ± 1.1%	82.7% ± 1.1%	81.4% ± 1.0%	85.5% ± 2.4%	86.1% ± 2.4%	73.7% ± 2.1%	68.3% ± 2.1%	58.9% ± 3.4%	68.7 ± 7.6	3
APPLE-Full	67.6% ± 4.0%	70.4% ± 3.5%	76.7% ± 4.0%	78.9% ± 2.4%	87.3% ± 4.1%	75.6% ± 9.3%	66.6% ± 3.2%	65.9% ± 5.7%	56.5% ± 4.7%	76.7 ± 1.5	3
APPLE-ALA	75.1% ± 0.1%	78.3% ± 0.3%	87.4% ± 1.4%	85.4% ± 1.5%	93.1% ± 1.1%	88.9% ± 1.3%	79.8% ± 0.5%	72.3% ± 2.0%	57.5% ± 0.9%	72.7 ± 8.1	3
APPLE-PostALA	75.4% ± 1.1%	78.0% ± 1.0%	87.3% ± 3.7%	85.8% ± 3.1%	90.4% ± 3.7%	89.0% ± 3.2%	78.1% ± 3.6%	71.7% ± 1.8%	60.8% ± 4.8%	74.0 ± 8.7	3
Local-Centralized Upper Bound	85.2% ± 1.9%	87.5% ± 1.0%	95.2% ± 0.2%	93.8% ± 0.8%	97.5% ± 1.6%	96.1% ± 2.7%	90.8% ± 2.3%	82.8% ± 2.9%	70.2% ± 6.1%	—	3

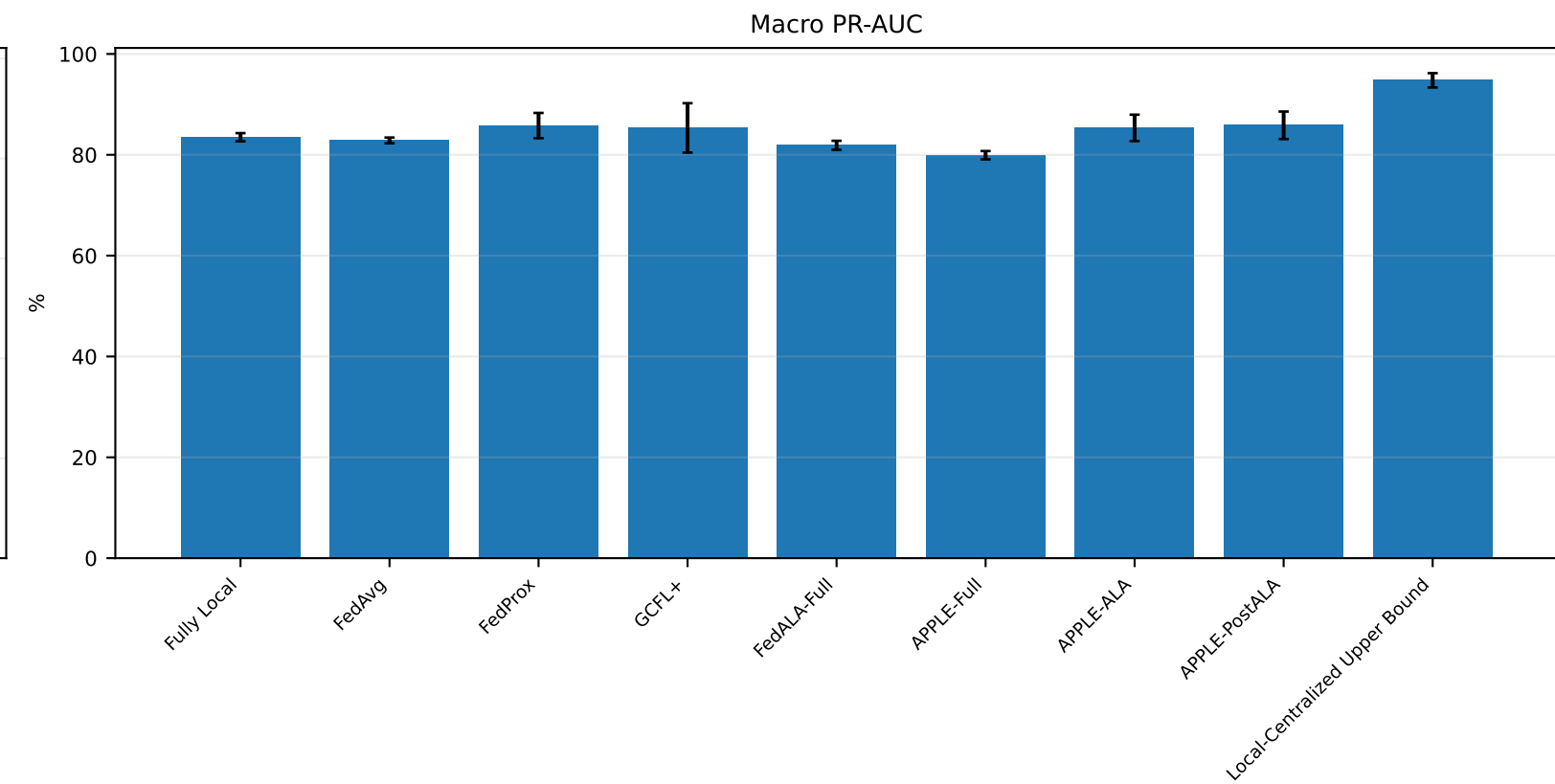
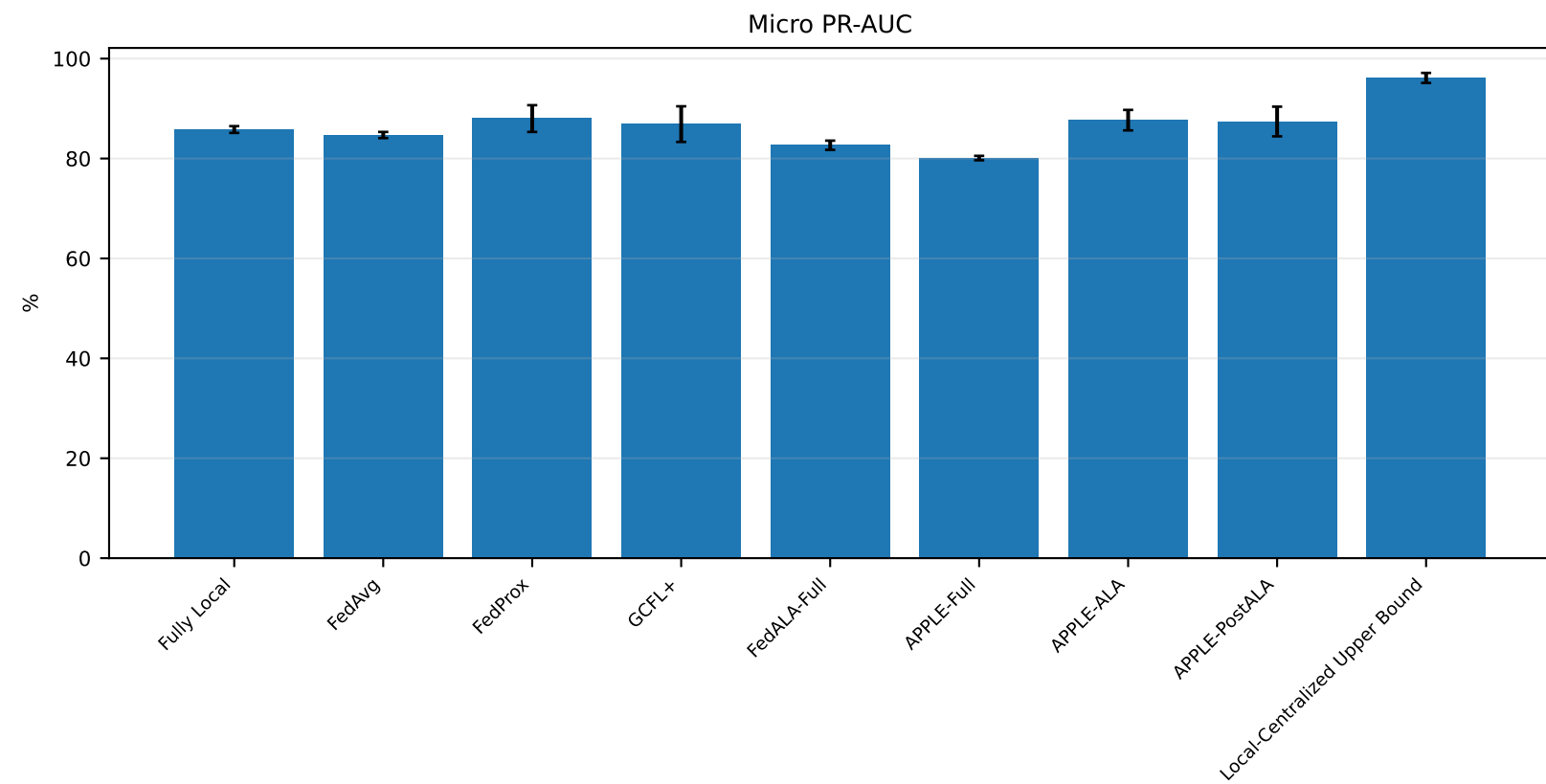
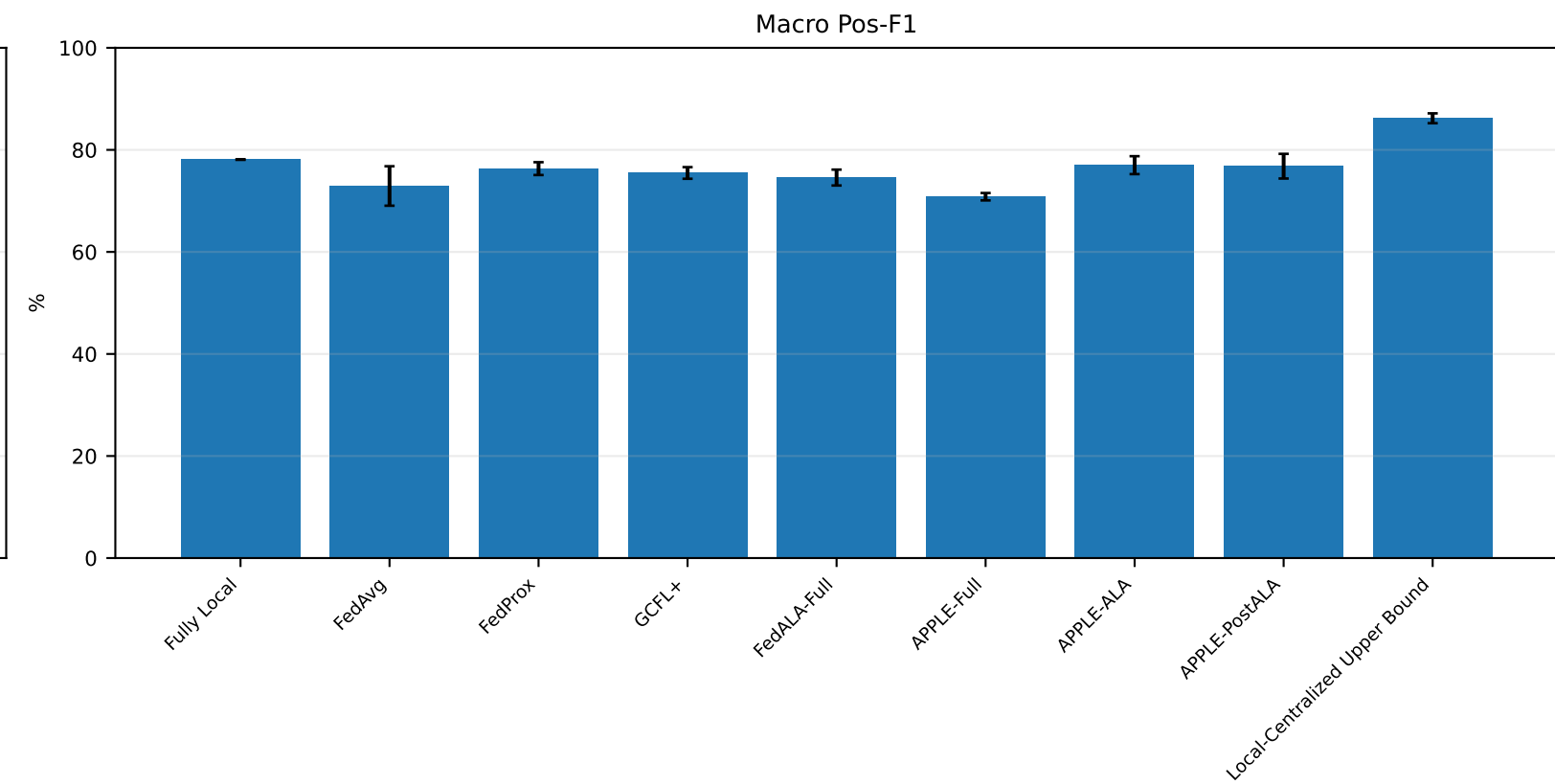
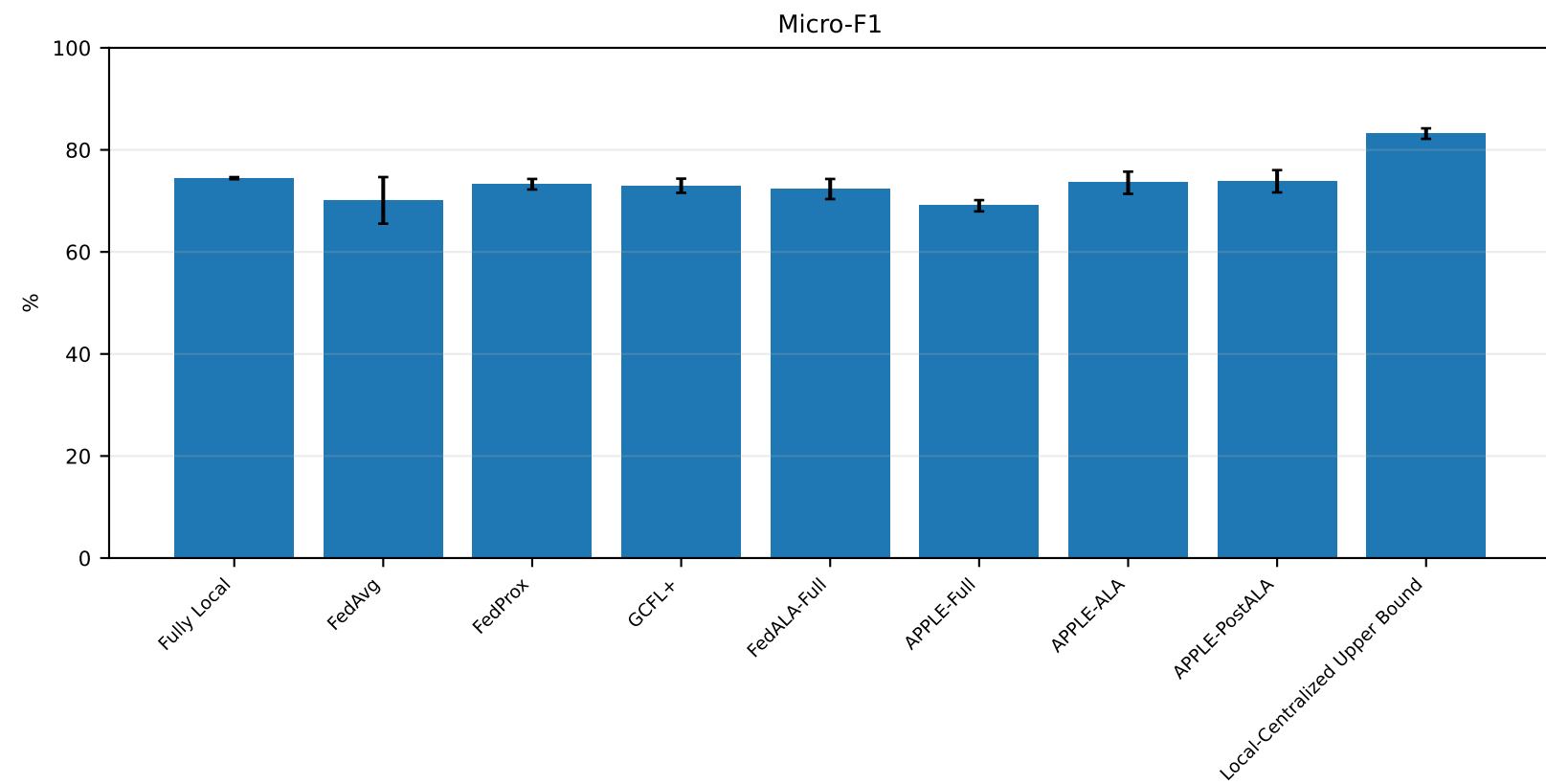
Main baselines + APPLE/FedALA/PostALA | IID / low heterogeneity ($q=0.2$) | selector=macro_pos_f1 | protocol=VO | overview bars
V+O: visible validation selection → oracle/full test



Main baselines + APPLE/FedALA/PostALA | IID / low heterogeneity (q=0.2) | selector=micro_pr_auc | protocol=OO
O+O: oracle/full validation selection → oracle/full test

Method	Micro-F1	Macro Pos-F1	Micro PR-AUC	Macro PR-AUC	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1	Sel. round	n
Fully Local	74.5% ± 0.2%	78.1% ± 0.1%	85.8% ± 0.7%	83.5% ± 0.8%	94.2% ± 0.8%	88.8% ± 0.3%	77.7% ± 0.4%	74.7% ± 0.5%	55.1% ± 0.5%	—	3
FedAvg	70.1% ± 4.6%	72.9% ± 3.9%	84.7% ± 0.6%	82.9% ± 0.6%	88.4% ± 3.6%	82.0% ± 5.8%	69.6% ± 5.5%	67.1% ± 5.1%	57.6% ± 5.2%	75.3 ± 4.7	3
FedProx	73.3% ± 1.0%	76.3% ± 1.3%	88.0% ± 2.7%	85.8% ± 2.5%	90.7% ± 3.9%	87.8% ± 0.8%	74.5% ± 3.9%	69.8% ± 2.7%	58.9% ± 0.8%	66.3 ± 10.0	3
GCFL+	73.0% ± 1.4%	75.5% ± 1.1%	86.9% ± 3.6%	85.3% ± 4.9%	86.6% ± 4.9%	87.0% ± 1.5%	72.8% ± 2.3%	71.0% ± 5.0%	59.9% ± 1.9%	72.0 ± 10.6	3
FedALA-Full	72.3% ± 2.0%	74.6% ± 1.6%	82.7% ± 0.9%	81.9% ± 0.9%	86.4% ± 1.3%	84.8% ± 3.9%	72.6% ± 0.7%	69.3% ± 3.3%	59.7% ± 3.1%	69.3 ± 4.0	3
APPLE-Full	69.0% ± 1.1%	70.8% ± 0.7%	80.1% ± 0.4%	79.9% ± 0.8%	85.1% ± 4.4%	73.1% ± 3.9%	68.7% ± 2.6%	68.0% ± 3.2%	59.3% ± 1.9%	71.0 ± 13.9	3
APPLE-ALA	73.6% ± 2.2%	77.0% ± 1.8%	87.7% ± 2.0%	85.3% ± 2.6%	91.3% ± 1.7%	88.7% ± 1.3%	79.7% ± 2.0%	70.2% ± 3.4%	55.2% ± 2.8%	77.0 ± 4.4	3
APPLE-PostALA	73.8% ± 2.2%	76.8% ± 2.4%	87.4% ± 3.0%	85.9% ± 2.7%	89.5% ± 4.1%	88.8% ± 2.9%	75.3% ± 5.8%	72.0% ± 2.4%	58.3% ± 0.3%	72.7 ± 6.8	3
Local-Centralized Upper Bound	83.2% ± 1.0%	86.2% ± 1.0%	96.1% ± 1.0%	94.8% ± 1.4%	98.0% ± 0.9%	96.5% ± 1.6%	89.8% ± 2.4%	81.0% ± 0.8%	65.7% ± 1.7%	—	3

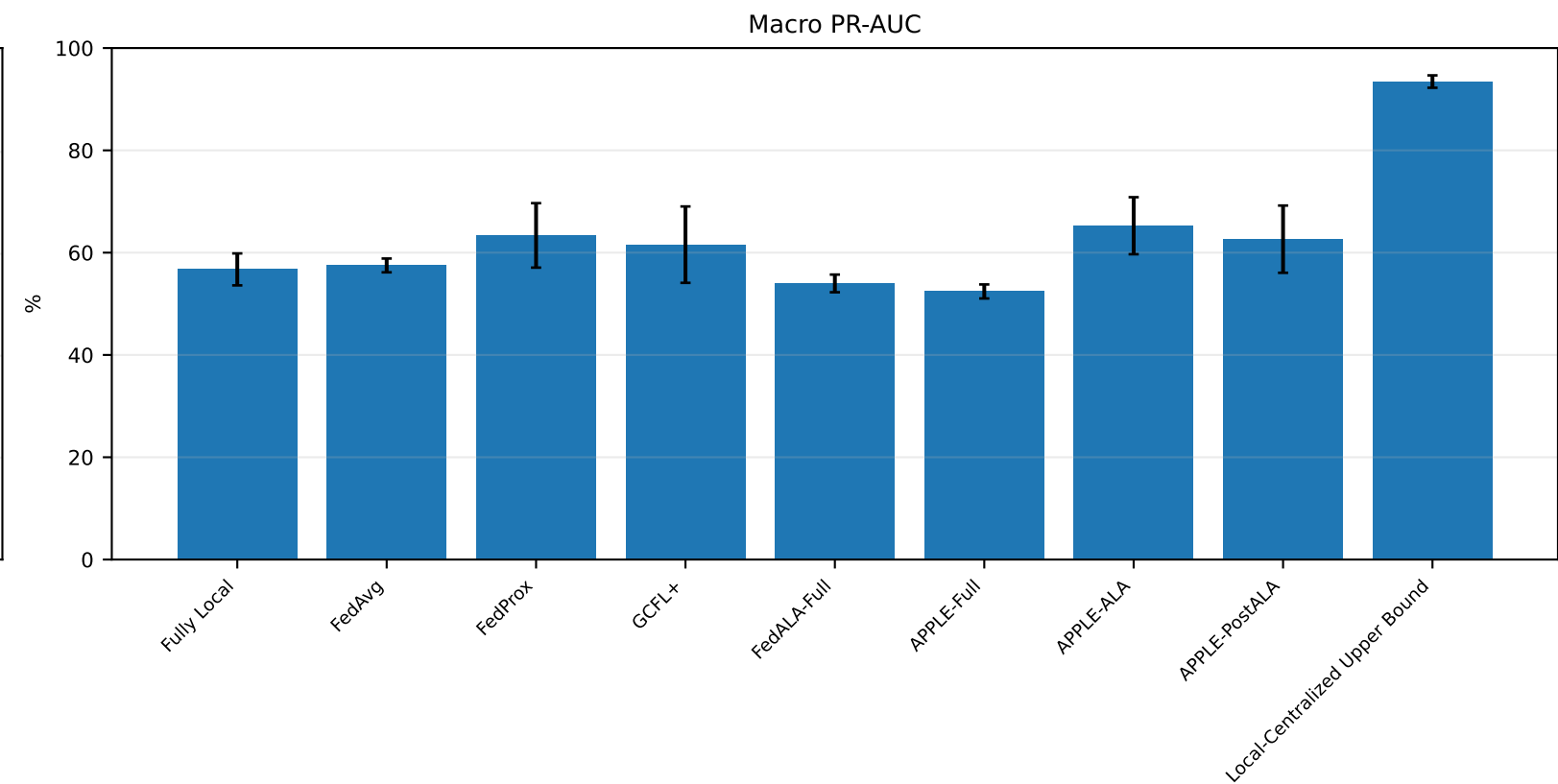
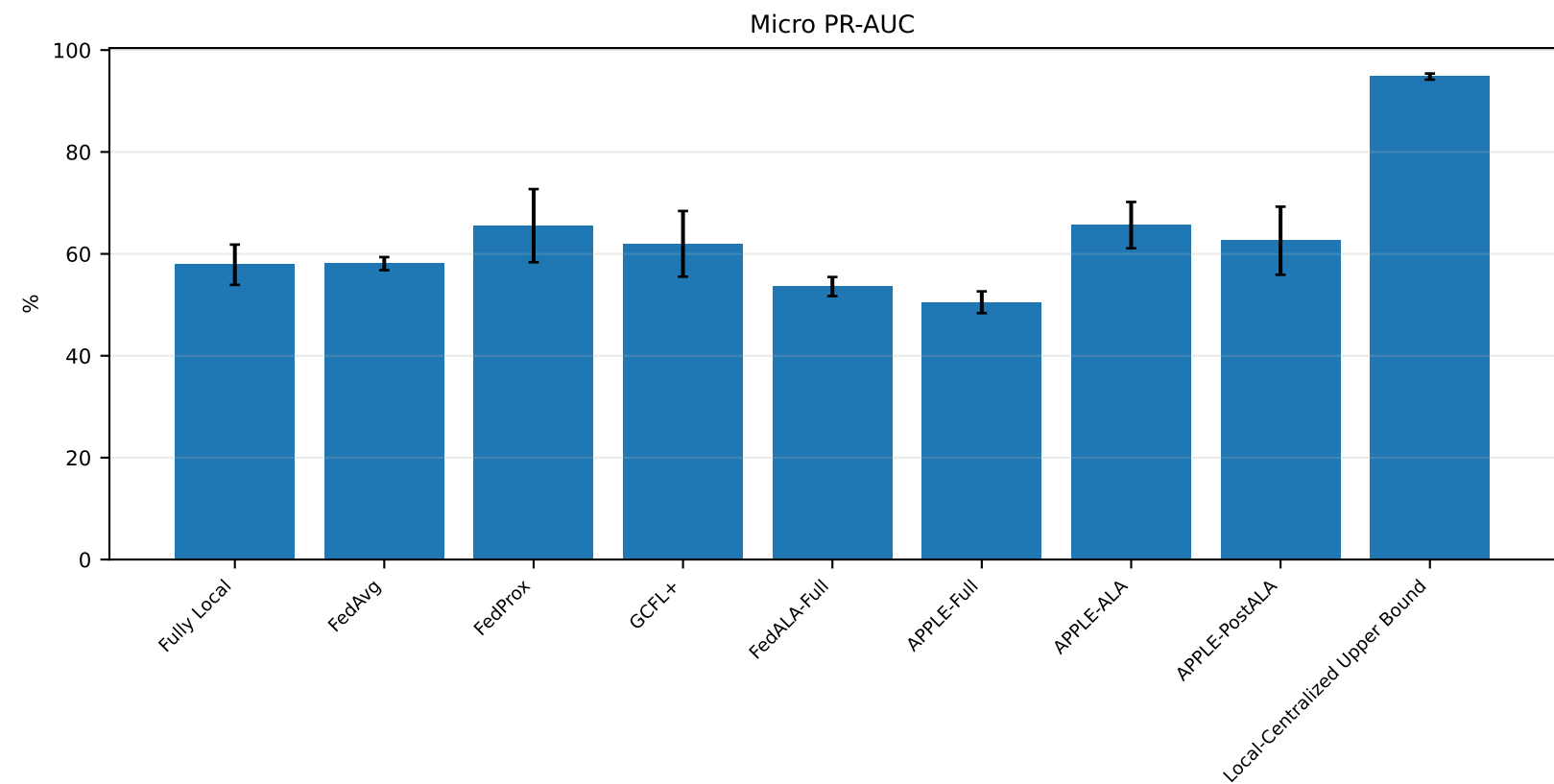
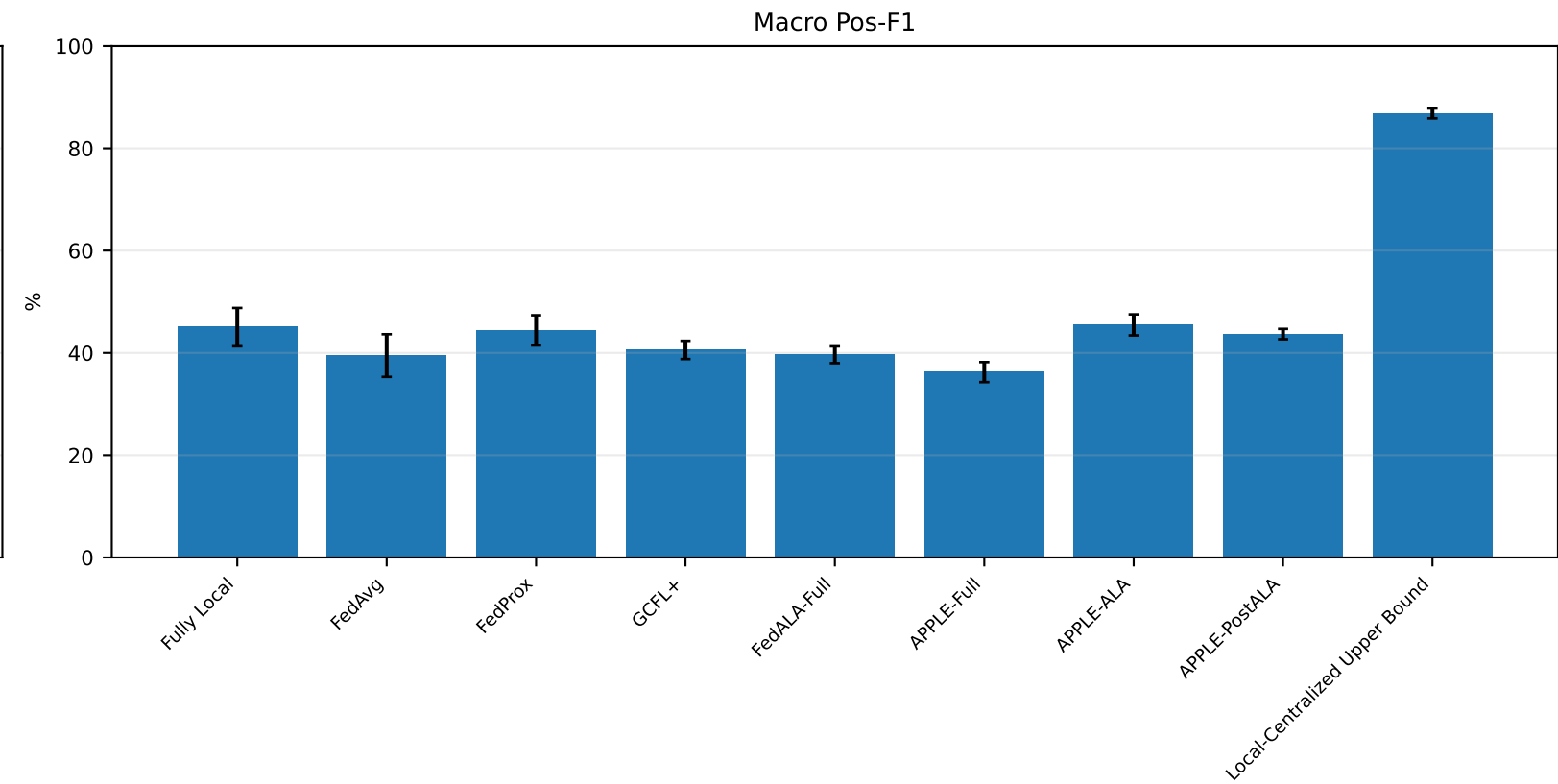
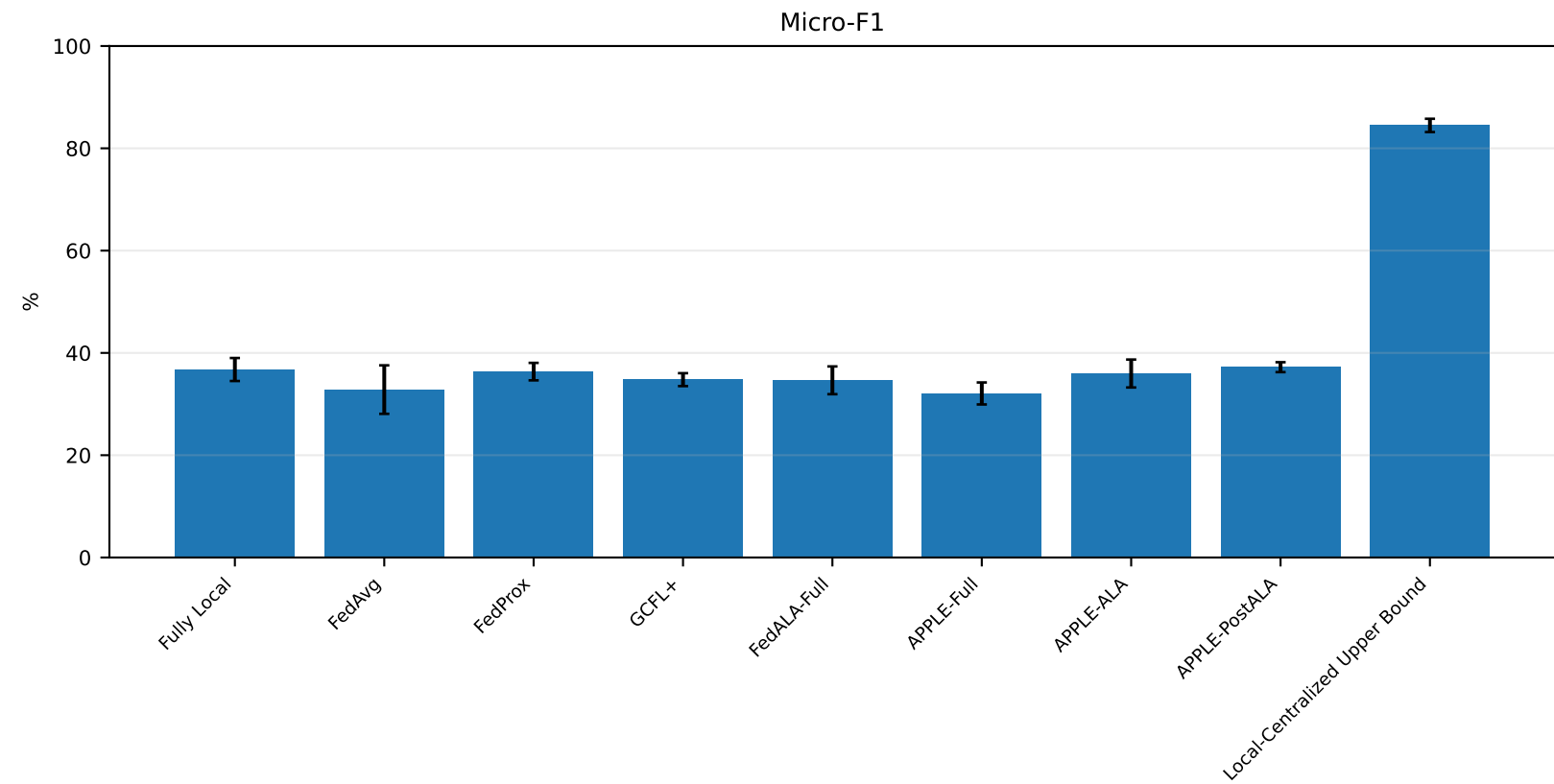
Main baselines + APPLE/FedALA/PostALA | IID / low heterogeneity ($q=0.2$) | selector=micro_pr_auc | protocol=OO | overview bars
O+O: oracle/full validation selection → oracle/full test



Main baselines + APPLE/FedALA/PostALA | IID / low heterogeneity (q=0.2) | selector=micro_pr_auc | protocol=VV
V+V: visible validation selection → visible test

Method	Micro-F1	Macro Pos-F1	Micro PR-AUC	Macro PR-AUC	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1	Sel. round	n
Fully Local	36.8% ± 2.2%	45.0% ± 3.7%	57.9% ± 4.0%	56.7% ± 3.1%	72.5% ± 5.3%	56.2% ± 9.0%	39.2% ± 2.2%	36.8% ± 2.8%	20.5% ± 0.5%	—	3
FedAvg	32.8% ± 4.7%	39.5% ± 4.2%	58.1% ± 1.3%	57.5% ± 1.3%	63.0% ± 9.9%	50.5% ± 9.8%	32.5% ± 5.9%	29.6% ± 4.8%	21.8% ± 3.5%	74.7 ± 4.0	3
FedProx	36.3% ± 1.7%	44.4% ± 2.9%	65.5% ± 7.2%	63.4% ± 6.3%	68.1% ± 10.7%	60.7% ± 2.2%	38.2% ± 4.4%	32.4% ± 2.8%	22.8% ± 1.6%	67.7 ± 9.3	3
GCFL+	34.8% ± 1.3%	40.6% ± 1.8%	62.0% ± 6.4%	61.6% ± 7.5%	55.7% ± 11.9%	56.5% ± 2.3%	34.7% ± 2.7%	33.1% ± 4.5%	22.9% ± 1.1%	72.0 ± 10.6	3
FedALA-Full	34.6% ± 2.7%	39.6% ± 1.6%	53.6% ± 1.9%	54.0% ± 1.7%	54.8% ± 1.2%	53.0% ± 3.1%	34.9% ± 1.9%	32.1% ± 3.9%	23.4% ± 3.7%	67.7 ± 6.7	3
APPLE-Full	32.1% ± 2.1%	36.2% ± 2.0%	50.5% ± 2.1%	52.4% ± 1.4%	58.1% ± 2.0%	37.4% ± 8.2%	32.6% ± 1.6%	29.7% ± 2.5%	23.3% ± 1.5%	70.3 ± 13.3	3
APPLE-ALA	36.0% ± 2.7%	45.5% ± 2.1%	65.6% ± 4.5%	65.3% ± 5.6%	73.0% ± 5.7%	60.2% ± 1.4%	42.1% ± 4.2%	31.7% ± 3.5%	20.3% ± 2.5%	79.0 ± 1.0	3
APPLE-PostALA	37.2% ± 1.0%	43.7% ± 1.0%	62.6% ± 6.7%	62.6% ± 6.6%	60.5% ± 4.4%	57.7% ± 2.4%	41.9% ± 4.4%	35.8% ± 0.5%	22.5% ± 3.4%	73.7 ± 2.5	3
Local-Centralized Upper Bound	84.5% ± 1.3%	86.8% ± 1.0%	94.8% ± 0.6%	93.4% ± 1.2%	97.9% ± 0.1%	95.8% ± 2.3%	88.5% ± 0.8%	82.4% ± 4.3%	69.5% ± 3.4%	—	3

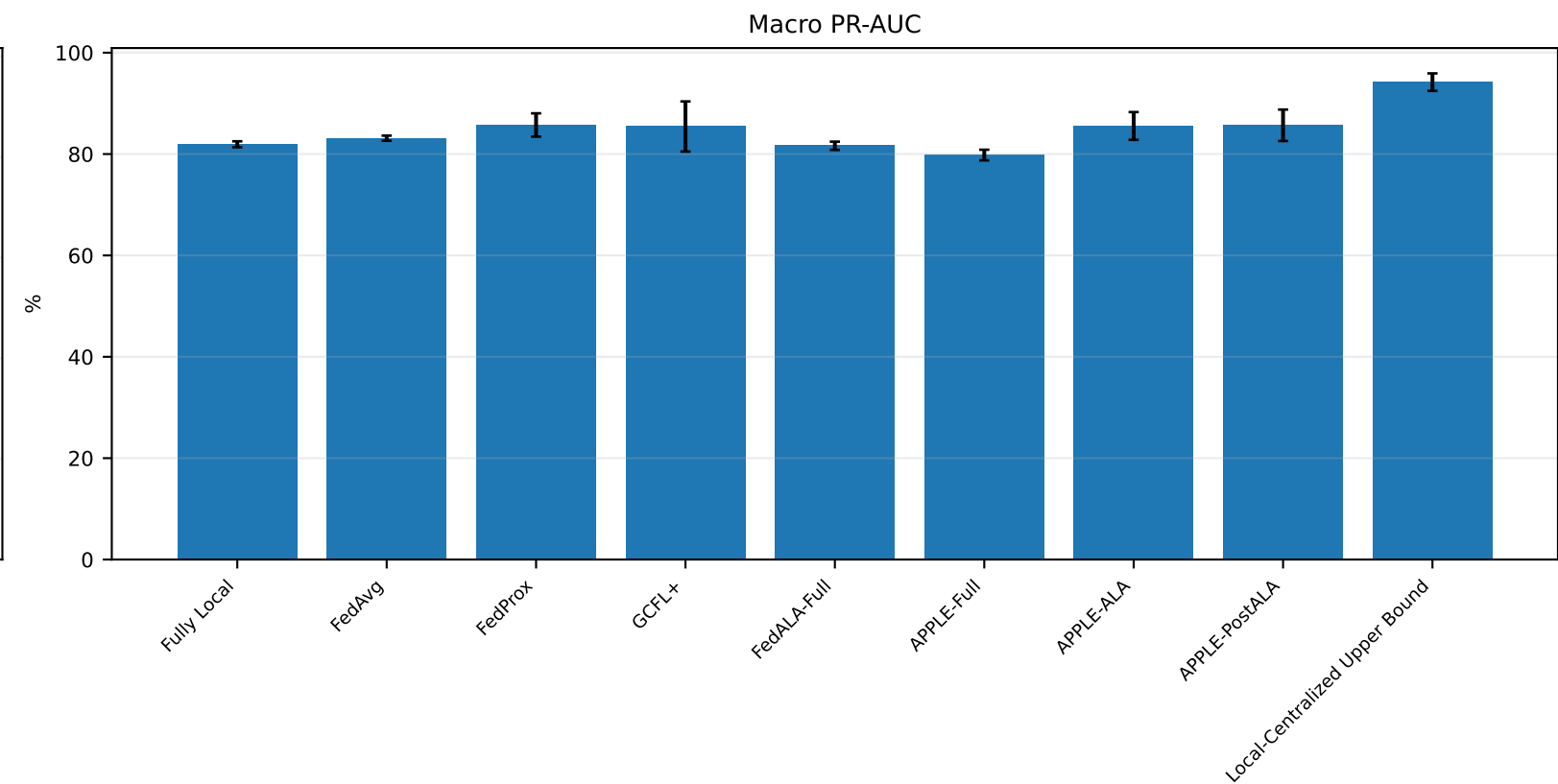
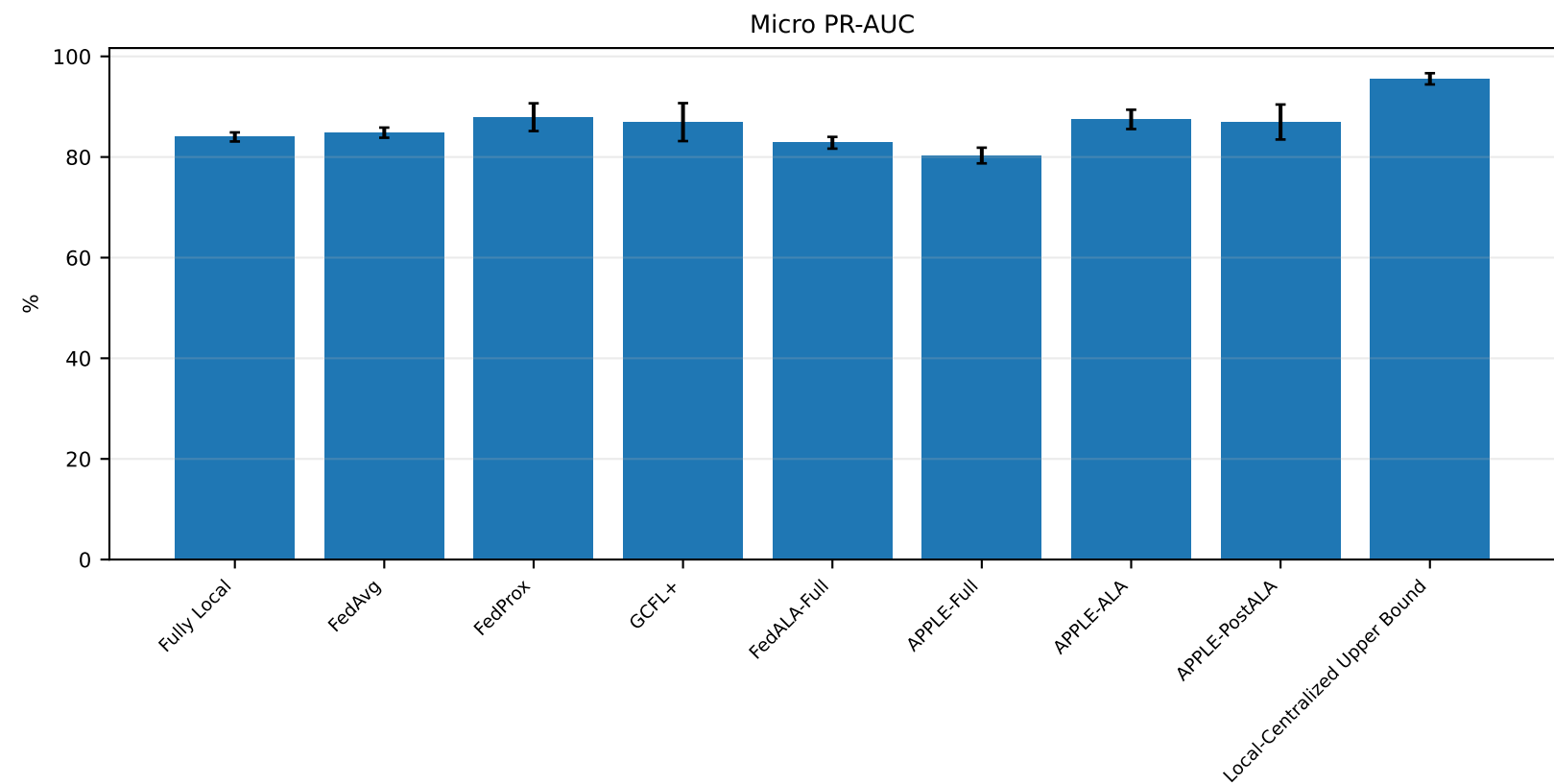
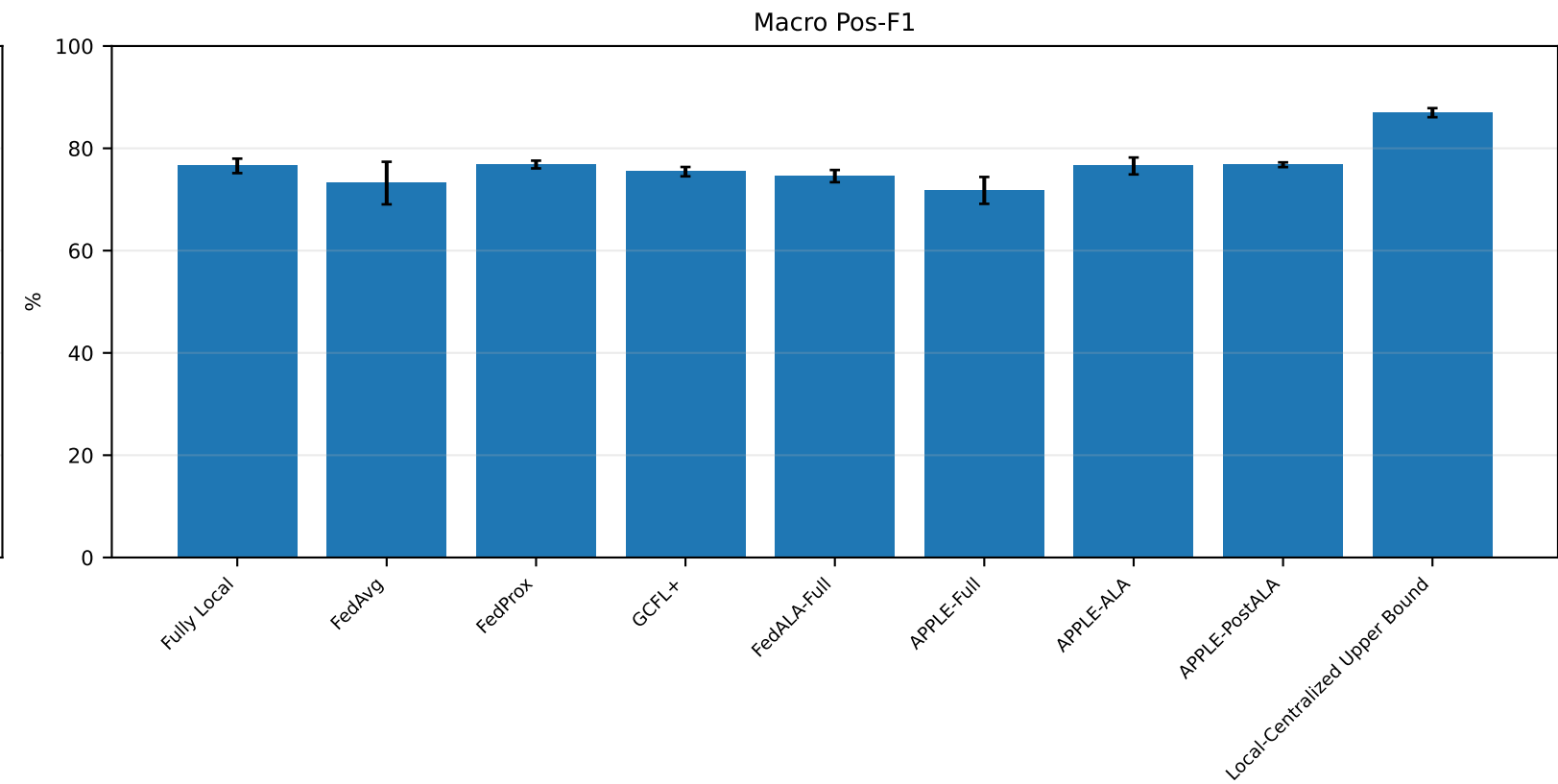
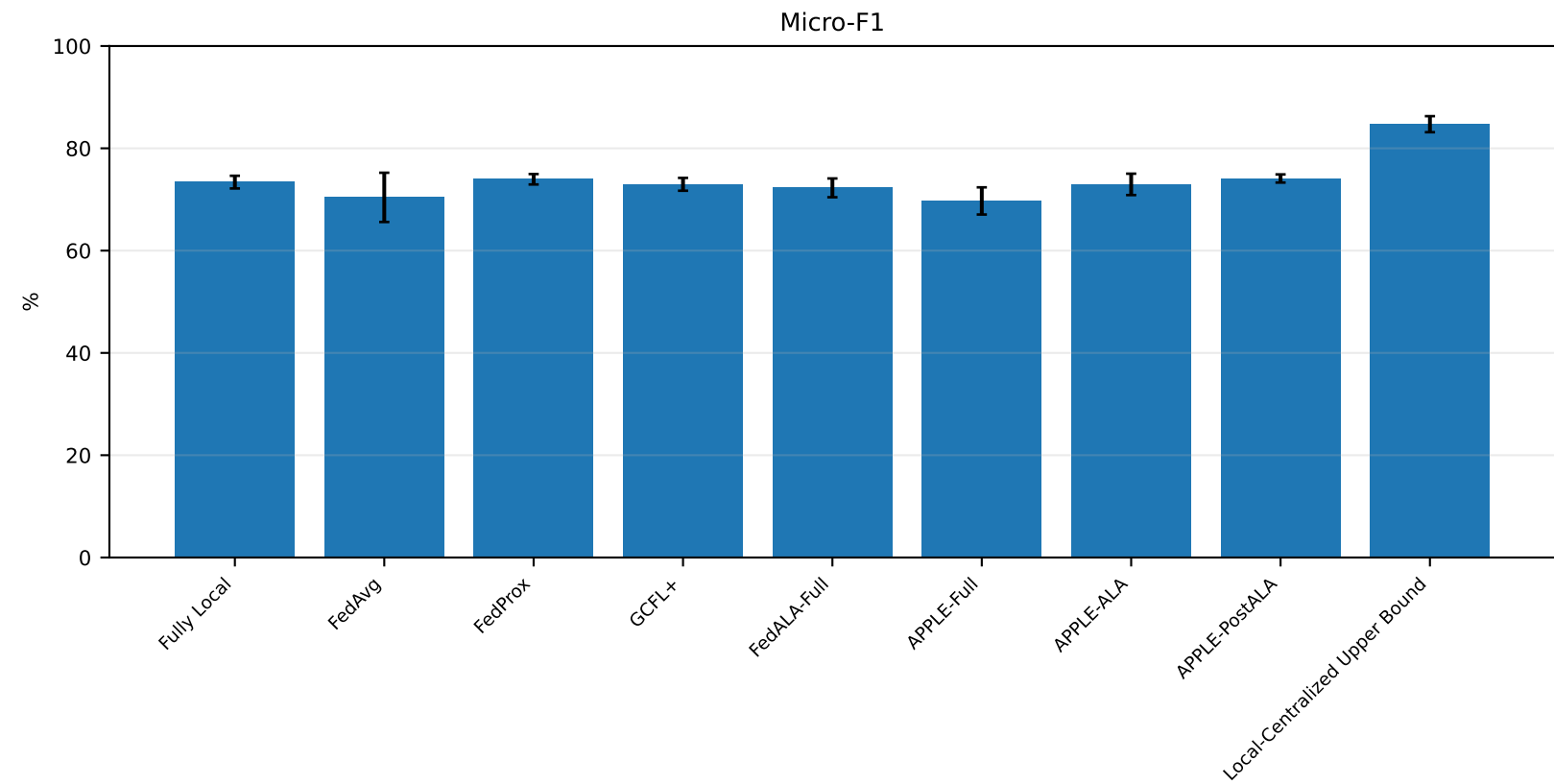
Main baselines + APPLE/FedALA/PostALA | IID / low heterogeneity ($q=0.2$) | selector=micro_pr_auc | protocol=VV | overview bars
V+V: visible validation selection → visible test



Main baselines + APPLE/FedALA/PostALA | IID / low heterogeneity (q=0.2) | selector=micro_pr_auc | protocol=VO
V+O: visible validation selection → oracle/full test

Method	Micro-F1	Macro Pos-F1	Micro PR-AUC	Macro PR-AUC	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1	Sel. round	n
Fully Local	73.4% ± 1.2%	76.6% ± 1.4%	84.0% ± 0.9%	81.9% ± 0.6%	91.7% ± 1.3%	86.8% ± 3.5%	75.5% ± 1.8%	74.1% ± 1.6%	54.8% ± 0.6%	—	3
FedAvg	70.4% ± 4.8%	73.2% ± 4.2%	84.8% ± 1.0%	83.1% ± 0.5%	88.4% ± 4.4%	82.5% ± 4.8%	70.2% ± 6.6%	67.0% ± 5.0%	58.0% ± 5.6%	74.7 ± 4.0	3
FedProx	74.0% ± 1.0%	76.8% ± 0.8%	87.9% ± 2.8%	85.7% ± 2.3%	90.7% ± 2.6%	88.0% ± 0.5%	75.5% ± 2.4%	70.0% ± 2.5%	59.8% ± 2.6%	67.7 ± 9.3	3
GCFL+	73.0% ± 1.3%	75.4% ± 0.9%	86.9% ± 3.8%	85.4% ± 4.9%	86.3% ± 4.1%	87.1% ± 1.8%	72.8% ± 2.1%	71.3% ± 4.7%	59.8% ± 1.7%	72.0 ± 10.6	3
FedALA-Full	72.3% ± 1.8%	74.6% ± 1.2%	82.8% ± 1.2%	81.6% ± 0.8%	85.3% ± 0.5%	85.3% ± 1.4%	72.5% ± 3.1%	69.9% ± 3.0%	59.8% ± 4.4%	67.7 ± 6.7	3
APPLE-Full	69.7% ± 2.7%	71.8% ± 2.6%	80.3% ± 1.6%	79.8% ± 1.0%	87.4% ± 0.9%	74.4% ± 7.1%	69.8% ± 1.2%	68.0% ± 3.5%	59.2% ± 2.2%	70.3 ± 13.3	3
APPLE-ALA	73.0% ± 2.1%	76.6% ± 1.7%	87.5% ± 1.9%	85.5% ± 2.7%	92.7% ± 1.2%	87.8% ± 0.9%	77.6% ± 2.6%	69.1% ± 3.0%	55.5% ± 2.9%	79.0 ± 1.0	3
APPLE-PostALA	74.1% ± 0.8%	76.8% ± 0.5%	87.0% ± 3.5%	85.7% ± 3.1%	87.3% ± 1.4%	87.4% ± 2.0%	77.7% ± 3.0%	73.2% ± 0.7%	58.2% ± 4.0%	73.7 ± 2.5	3
Local-Centralized Upper Bound	84.7% ± 1.6%	87.0% ± 0.9%	95.5% ± 1.1%	94.2% ± 1.7%	96.9% ± 0.9%	96.3% ± 1.7%	89.2% ± 0.8%	82.3% ± 3.6%	70.2% ± 4.0%	—	3

Main baselines + APPLE/FedALA/PostALA | IID / low heterogeneity ($q=0.2$) | selector=micro_pr_auc | protocol=VO | overview bars
V+O: visible validation selection → oracle/full test

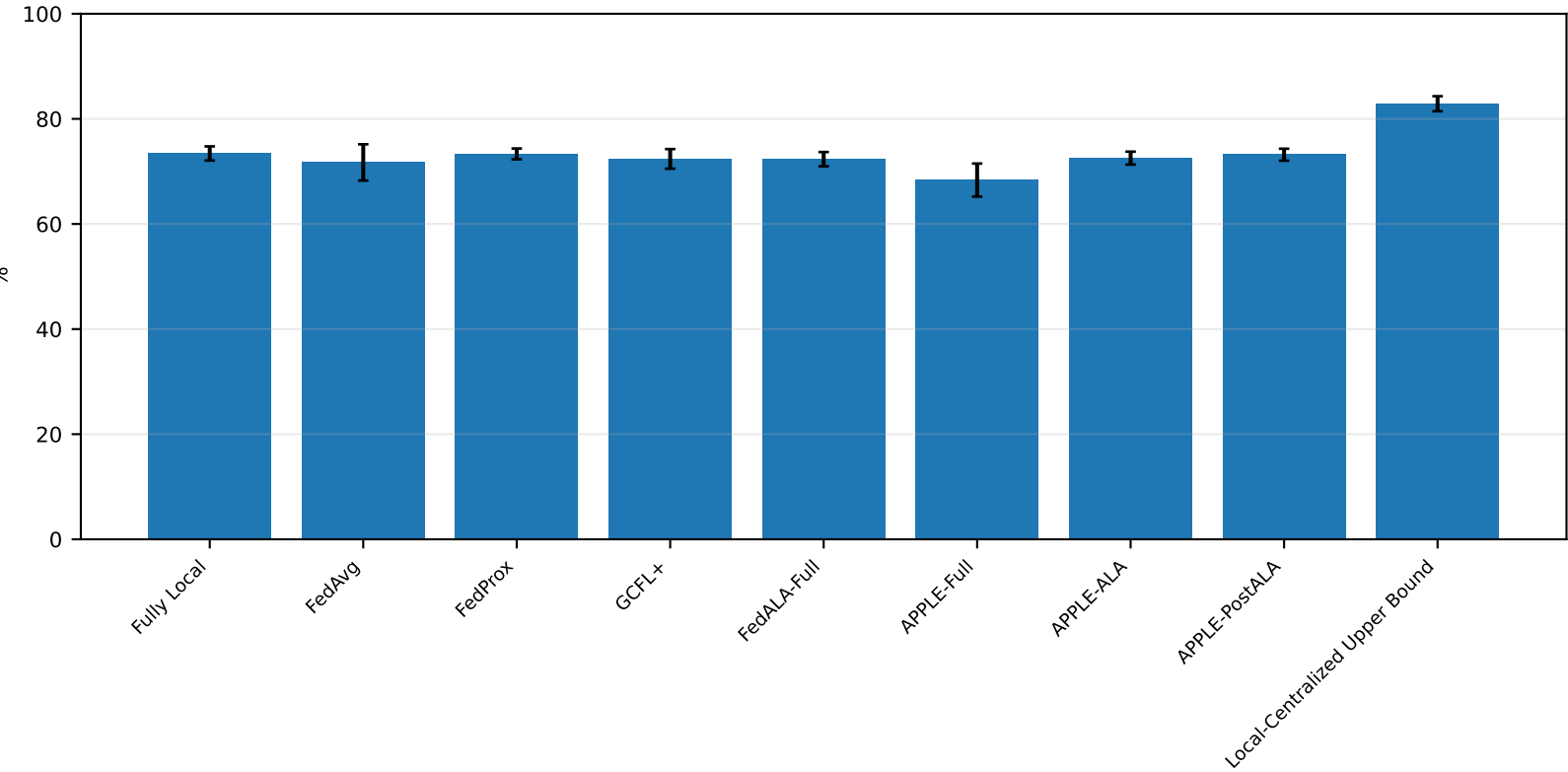


Main baselines + APPLE/FedALA/PostALA | IID / low heterogeneity (q=0.2) | selector=macro_pr_auc | protocol=OO
O+O: oracle/full validation selection → oracle/full test

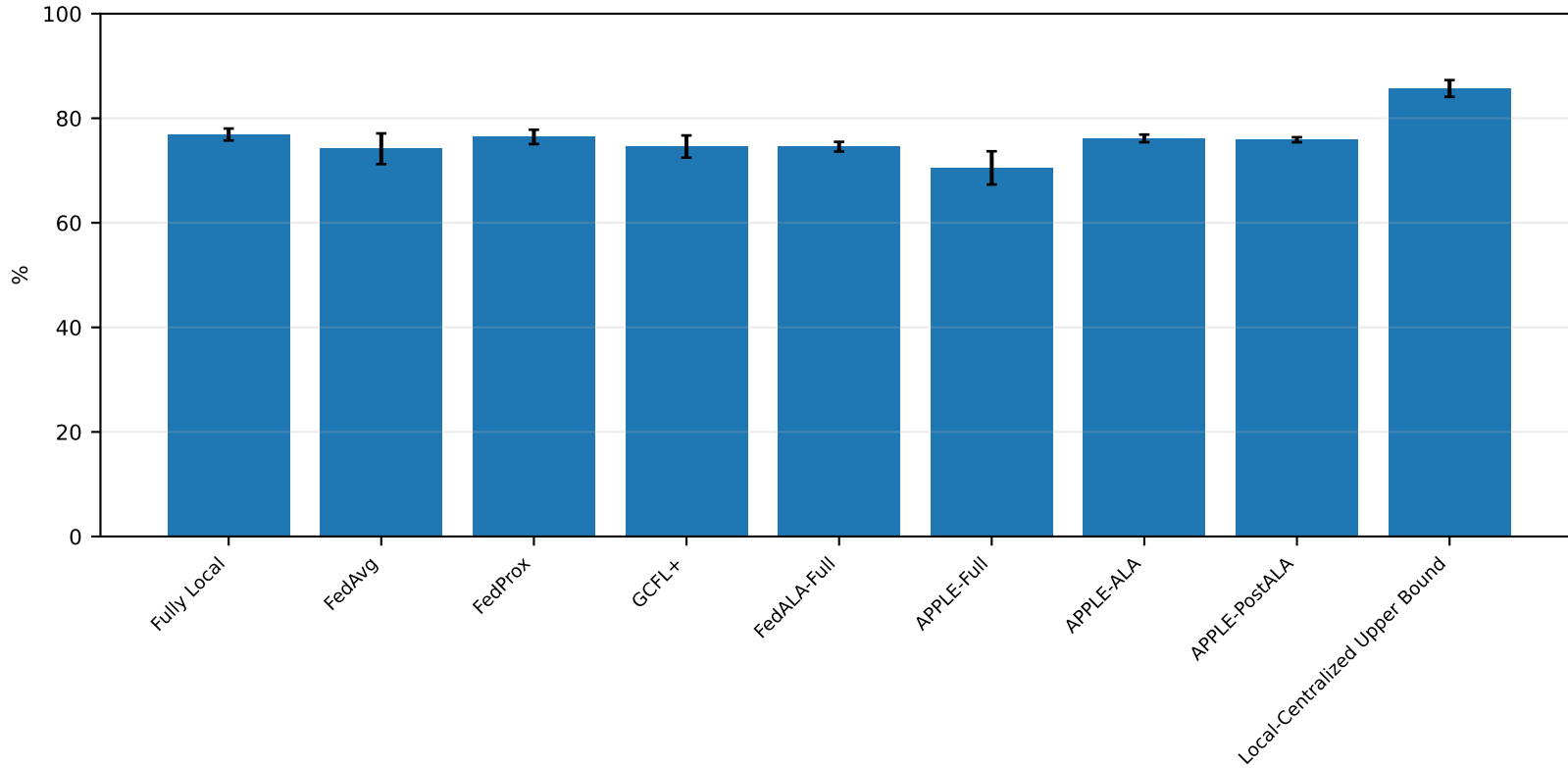
Method	Micro-F1	Macro Pos-F1	Micro PR-AUC	Macro PR-AUC	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1	Sel. round	n
Fully Local	73.4% ± 1.3%	76.9% ± 1.1%	84.7% ± 2.2%	82.8% ± 2.2%	92.1% ± 0.9%	87.1% ± 1.3%	76.2% ± 1.3%	74.0% ± 1.1%	55.1% ± 2.1%	—	3
FedAvg	71.7% ± 3.5%	74.2% ± 2.9%	85.6% ± 0.4%	83.8% ± 0.2%	87.5% ± 3.4%	83.0% ± 5.7%	72.6% ± 2.3%	68.0% ± 4.3%	59.7% ± 4.5%	78.3 ± 1.5	3
FedProx	73.3% ± 1.0%	76.4% ± 1.4%	87.9% ± 2.5%	85.9% ± 2.0%	92.3% ± 2.7%	87.2% ± 1.3%	73.2% ± 5.3%	69.9% ± 2.5%	59.6% ± 0.5%	74.7 ± 4.2	3
GCFL+	72.4% ± 1.9%	74.6% ± 2.1%	87.3% ± 3.3%	86.1% ± 3.9%	84.2% ± 7.6%	85.9% ± 0.5%	71.0% ± 0.7%	71.9% ± 3.3%	59.9% ± 1.5%	73.7 ± 7.8	3
FedALA-Full	72.3% ± 1.4%	74.6% ± 0.9%	82.7% ± 0.8%	81.8% ± 0.9%	86.4% ± 1.6%	85.1% ± 2.8%	72.2% ± 0.3%	69.5% ± 2.5%	59.7% ± 2.6%	69.3 ± 4.0	3
APPLE-Full	68.4% ± 3.1%	70.5% ± 3.2%	79.0% ± 0.7%	80.0% ± 1.0%	86.7% ± 3.0%	71.7% ± 8.5%	68.5% ± 2.9%	67.5% ± 2.9%	58.2% ± 3.6%	71.0 ± 13.9	3
APPLE-ALA	72.5% ± 1.2%	76.1% ± 0.7%	87.3% ± 2.2%	85.5% ± 2.7%	92.2% ± 1.2%	87.9% ± 1.7%	76.7% ± 1.1%	68.4% ± 2.1%	55.6% ± 3.4%	78.3 ± 0.6	3
APPLE-PostALA	73.2% ± 1.1%	75.9% ± 0.5%	86.7% ± 3.1%	85.5% ± 2.9%	88.1% ± 6.0%	87.0% ± 1.3%	75.7% ± 2.3%	70.0% ± 1.7%	58.7% ± 3.8%	69.7 ± 11.1	3
Local-Centralized Upper Bound	82.9% ± 1.4%	85.7% ± 1.6%	95.4% ± 1.3%	94.2% ± 0.9%	97.5% ± 1.6%	95.4% ± 2.7%	88.8% ± 3.6%	80.5% ± 0.9%	66.2% ± 1.2%	—	3

Main baselines + APPLE/FedALA/PostALA | IID / low heterogeneity ($q=0.2$) | selector=macro_pr_auc | protocol=OO | overview bars
O+O: oracle/full validation selection → oracle/full test

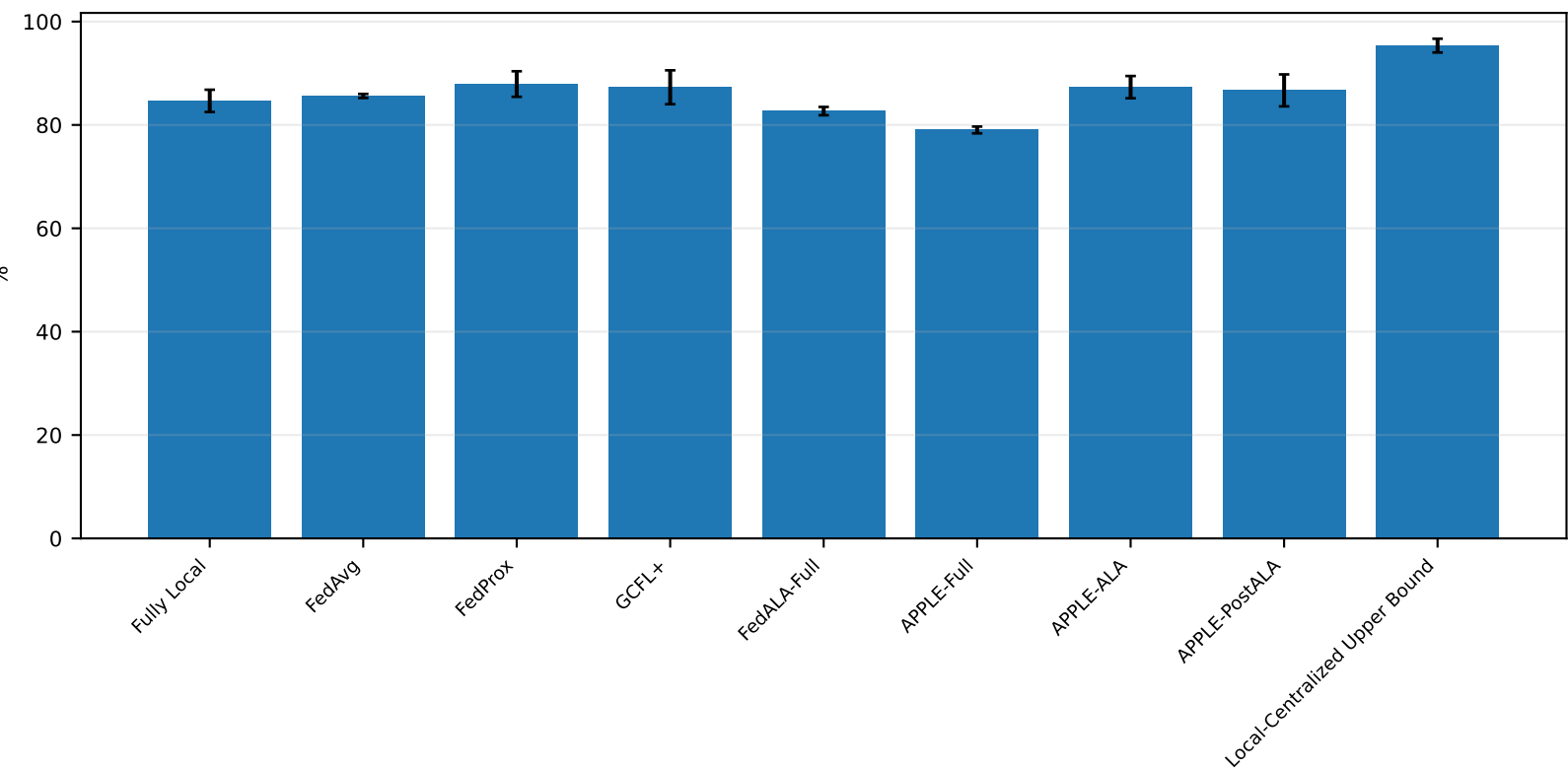
Micro-F1



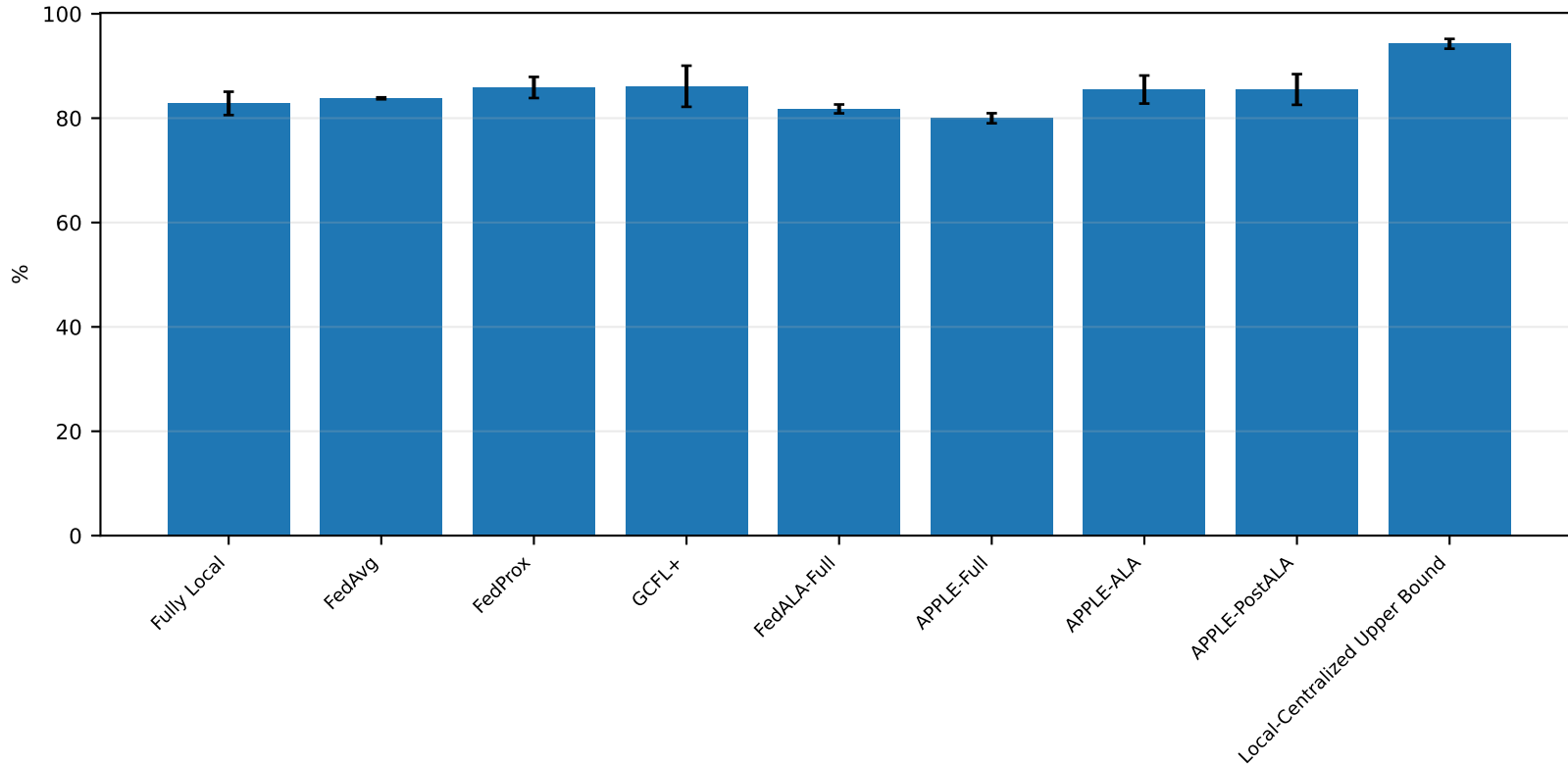
Macro Pos-F1



Micro PR-AUC



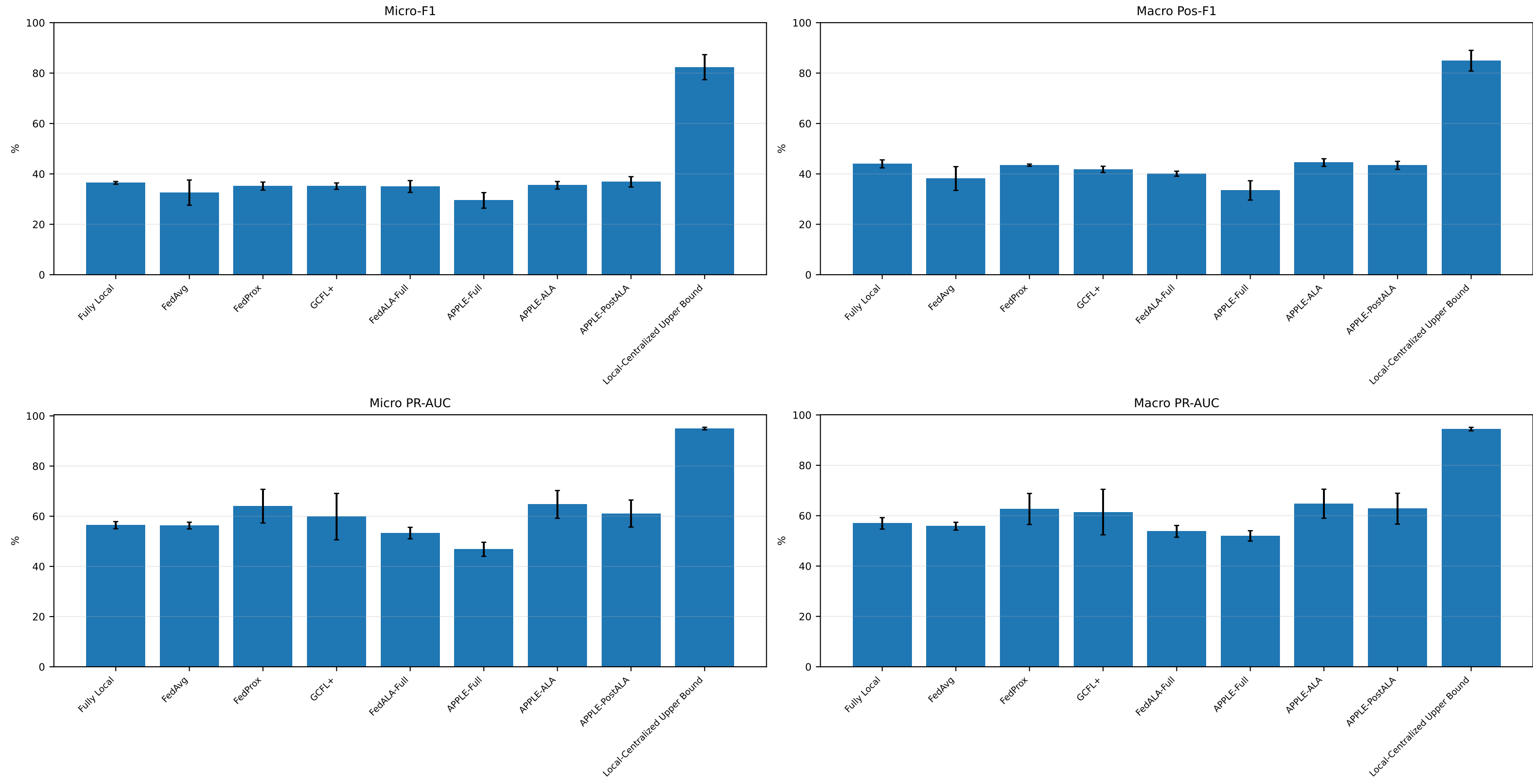
Macro PR-AUC



Main baselines + APPLE/FedALA/PostALA | IID / low heterogeneity (q=0.2) | selector=macro_pr_auc | protocol=VV
V+V: visible validation selection → visible test

Method	Micro-F1	Macro Pos-F1	Micro PR-AUC	Macro PR-AUC	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1	Sel. round	n
Fully Local	36.5% ± 0.5%	44.0% ± 1.6%	56.4% ± 1.4%	57.0% ± 2.3%	70.6% ± 3.0%	53.3% ± 3.9%	38.2% ± 1.4%	36.9% ± 2.3%	21.0% ± 0.3%	—	3
FedAvg	32.6% ± 5.0%	38.2% ± 4.7%	56.3% ± 1.3%	55.8% ± 1.5%	58.3% ± 4.1%	49.1% ± 10.4%	31.7% ± 6.0%	29.7% ± 4.6%	22.1% ± 3.5%	74.0 ± 3.6	3
FedProx	35.2% ± 1.6%	43.5% ± 0.4%	64.0% ± 6.7%	62.7% ± 6.2%	67.4% ± 6.3%	58.0% ± 3.6%	39.8% ± 4.4%	30.2% ± 3.6%	21.9% ± 2.5%	68.0 ± 11.0	3
GCFL+	35.1% ± 1.3%	41.8% ± 1.2%	59.8% ± 9.2%	61.4% ± 9.0%	59.9% ± 7.0%	58.3% ± 2.9%	34.3% ± 1.2%	33.7% ± 4.4%	22.7% ± 1.6%	77.7 ± 1.5	3
FedALA-Full	35.0% ± 2.3%	40.1% ± 1.0%	53.3% ± 2.3%	53.8% ± 2.3%	55.3% ± 2.2%	53.1% ± 2.2%	36.1% ± 2.6%	32.7% ± 4.1%	23.3% ± 3.4%	67.7 ± 6.7	3
APPLE-Full	29.5% ± 3.1%	33.5% ± 3.8%	46.8% ± 2.8%	52.0% ± 2.1%	53.1% ± 8.8%	35.5% ± 8.9%	28.7% ± 1.1%	28.5% ± 3.7%	21.5% ± 2.1%	63.7 ± 9.1	3
APPLE-ALA	35.5% ± 1.5%	44.5% ± 1.5%	64.7% ± 5.5%	64.7% ± 5.8%	71.3% ± 7.5%	59.6% ± 1.7%	40.1% ± 1.6%	30.9% ± 2.2%	20.6% ± 2.5%	78.3 ± 0.6	3
APPLE-PostALA	36.8% ± 2.1%	43.4% ± 1.6%	61.1% ± 5.4%	62.8% ± 6.1%	61.3% ± 17.4%	60.3% ± 7.4%	40.0% ± 1.1%	32.0% ± 0.7%	23.4% ± 4.3%	76.7 ± 5.8	3
Local-Centralized Upper Bound	82.3% ± 5.0%	84.9% ± 4.1%	94.9% ± 0.5%	94.4% ± 0.7%	97.1% ± 1.2%	95.6% ± 0.7%	83.6% ± 10.7%	80.0% ± 3.7%	68.2% ± 6.2%	—	3

Main baselines + APPLE/FedALA/PostALA | IID / low heterogeneity ($q=0.2$) | selector=macro_pr_auc | protocol=VV | overview bars
V+V: visible validation selection → visible test



Main baselines + APPLE/FedALA/PostALA | IID / low heterogeneity (q=0.2) | selector=macro_pr_auc | protocol=VO
V+O: visible validation selection → oracle/full test

Method	Micro-F1	Macro Pos-F1	Micro PR-AUC	Macro PR-AUC	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1	Sel. round	n
Fully Local	73.3% ± 0.7%	76.5% ± 1.0%	85.0% ± 0.9%	83.3% ± 1.0%	92.9% ± 1.3%	85.2% ± 2.5%	75.7% ± 1.0%	73.9% ± 1.7%	54.7% ± 0.9%	—	3
FedAvg	70.8% ± 4.3%	73.5% ± 3.5%	85.5% ± 0.9%	83.9% ± 0.2%	88.1% ± 2.9%	83.0% ± 4.7%	70.2% ± 5.7%	68.0% ± 4.3%	58.4% ± 5.0%	74.0 ± 3.6	3
FedProx	73.2% ± 1.0%	76.6% ± 0.1%	88.2% ± 2.9%	86.3% ± 1.9%	91.3% ± 4.1%	87.9% ± 0.8%	77.0% ± 3.4%	68.3% ± 3.4%	58.4% ± 3.4%	68.0 ± 11.0	3
GCFL+	73.0% ± 1.2%	75.6% ± 0.7%	86.3% ± 3.6%	85.6% ± 4.2%	87.3% ± 3.2%	86.7% ± 1.1%	72.5% ± 1.4%	71.8% ± 4.5%	59.5% ± 2.4%	77.7 ± 1.5	3
FedALA-Full	72.4% ± 2.2%	74.7% ± 1.6%	83.2% ± 1.7%	82.0% ± 1.5%	86.0% ± 0.5%	85.2% ± 0.5%	72.5% ± 2.3%	70.0% ± 3.6%	59.7% ± 4.4%	67.7 ± 6.7	3
APPLE-Full	67.3% ± 3.1%	69.4% ± 3.3%	78.9% ± 1.2%	80.4% ± 0.3%	83.9% ± 5.2%	72.7% ± 8.3%	67.2% ± 1.2%	66.1% ± 3.8%	57.0% ± 2.6%	63.7 ± 9.1	3
APPLE-ALA	72.5% ± 1.4%	76.1% ± 0.9%	87.5% ± 1.8%	85.8% ± 1.9%	92.4% ± 1.3%	87.1% ± 1.6%	76.7% ± 1.6%	68.5% ± 2.4%	55.7% ± 3.3%	78.3 ± 0.6	3
APPLE-PostALA	73.8% ± 1.6%	76.3% ± 0.6%	86.5% ± 2.9%	85.6% ± 3.2%	86.9% ± 6.0%	88.4% ± 3.7%	76.5% ± 0.3%	69.9% ± 0.7%	59.7% ± 5.5%	76.7 ± 5.8	3
Local-Centralized Upper Bound	82.4% ± 4.2%	85.0% ± 3.5%	95.2% ± 0.5%	94.4% ± 0.8%	98.2% ± 1.5%	94.8% ± 0.4%	82.7% ± 10.2%	81.5% ± 2.4%	68.0% ± 4.8%	—	3

Main baselines + APPLE/FedALA/PostALA | IID / low heterogeneity ($q=0.2$) | selector=macro_pr_auc | protocol=VO | overview bars
V+O: visible validation selection → oracle/full test

